

TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

- If $\lim_{n \rightarrow \infty} a_n = L$, then $\lim_{n \rightarrow \infty} a_{2n+1} = L$.
- $0.99999 \dots = 1$
- $\lim_{x \rightarrow 4} \left(\frac{2x}{x-4} - \frac{8}{x-4} \right) = \lim_{x \rightarrow 4} \frac{2x}{x-4} - \lim_{x \rightarrow 4} \frac{8}{x-4}$
- $\lim_{x \rightarrow 1} \frac{x^2 + 6x - 7}{x^2 + 5x - 6} = \frac{\lim_{x \rightarrow 1} (x^2 + 6x - 7)}{\lim_{x \rightarrow 1} (x^2 + 5x - 6)}$
- $\lim_{x \rightarrow 1} \frac{x-3}{x^2 + 2x - 4} = \frac{\lim_{x \rightarrow 1} (x-3)}{\lim_{x \rightarrow 1} (x^2 + 2x - 4)}$
- If $\lim_{x \rightarrow 5} f(x) = 2$ and $\lim_{x \rightarrow 5} g(x) = 0$, then $\lim_{x \rightarrow 5} [f(x)/g(x)]$ does not exist.
- If $\lim_{x \rightarrow 5} f(x) = 0$ and $\lim_{x \rightarrow 5} g(x) = 0$, then $\lim_{x \rightarrow 5} [f(x)/g(x)]$ does not exist.
- If $\lim_{x \rightarrow 6} [f(x)g(x)]$ exists, then the limit must be $f(6)g(6)$.
- If p is a polynomial, then $\lim_{x \rightarrow b} p(x) = p(b)$.
- If $\lim_{x \rightarrow 0} f(x) = \infty$ and $\lim_{x \rightarrow 0} g(x) = \infty$, then $\lim_{x \rightarrow 0} [f(x) - g(x)] = 0$.
- A function can have two different horizontal asymptotes.
- If f has domain $[0, \infty)$ and has no horizontal asymptote, then $\lim_{x \rightarrow \infty} f(x) = \infty$ or $\lim_{x \rightarrow \infty} f(x) = -\infty$.
- If the line $x = 1$ is a vertical asymptote of $y = f(x)$, then f is not defined at 1.
- If $f(1) > 0$ and $f(3) < 0$, then there exists a number c between 1 and 3 such that $f(c) = 0$.
- If f is continuous at 5 and $f(5) = 2$ and $f(4) = 3$, then $\lim_{x \rightarrow 2} f(4x^2 - 11) = 2$.
- If f is continuous on $[-1, 1]$ and $f(-1) = 4$ and $f(1) = 3$, then there exists a number r such that $|r| < 1$ and $f(r) = \pi$.

EXERCISES

1–4 Determine whether the sequence is convergent or divergent. If it is convergent, find its limit.

- $a_n = \frac{2 + n^3}{1 + 2n^3}$
- $a_n = \frac{9^{n+1}}{10^n}$
- $a_n = \frac{n^3}{1 + n^2}$
- $a_n = (-2)^n$

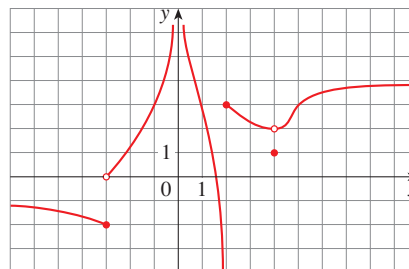
- Calculate the first eight terms of the sequence defined by $a_1 = 1$, $a_{n+1} = \frac{1}{3}a_n + 3$. Does it appear to be convergent? Assuming the limit exists, find its exact value.

- Drug concentration** A patient is injected with a drug at the same time every day. Before each injection, the concentration of the drug has dropped to 20% of its original value and the new dose raises the concentration by 0.25 mg/mL.
 - What is the concentration after four doses?
 - If C_n is the concentration after n doses, write a difference equation that expresses C_{n+1} in terms of C_n .
 - Solve the difference equation to find a formula for C_n .
 - Find the limiting value of the concentration.
- Express the repeating decimal $1.2345345345 \dots$ as a fraction.
- Logistic equation** Plot enough terms of the logistic difference equation to see the behavior of the terms. If the

sequence appears to be convergent, estimate its limit and then, assuming the limit exists, find its exact value. If not, describe how the terms behave.

- $x_{t+1} = 2.5x_t(1 - x_t)$, $x_0 = 0.5$
- $x_{t+1} = 3.3x_t(1 - x_t)$, $x_0 = 0.4$

- The graph of f is given.
 - Find each limit, or explain why it does not exist.
 - $\lim_{x \rightarrow 2^+} f(x)$
 - $\lim_{x \rightarrow -3^+} f(x)$
 - $\lim_{x \rightarrow -3} f(x)$
 - $\lim_{x \rightarrow 4} f(x)$
 - $\lim_{x \rightarrow 0} f(x)$
 - $\lim_{x \rightarrow 2^-} f(x)$
 - $\lim_{x \rightarrow \infty} f(x)$
 - $\lim_{x \rightarrow -\infty} f(x)$
 - State the equations of the horizontal asymptotes.
 - State the equations of the vertical asymptotes.
 - At what numbers is f discontinuous? Explain.



10. Sketch the graph of an example of a function f that satisfies all of the following conditions:

$$\lim_{x \rightarrow -\infty} f(x) = -2, \quad \lim_{x \rightarrow \infty} f(x) = 0, \quad \lim_{x \rightarrow -3} f(x) = \infty,$$

$$\lim_{x \rightarrow 3^-} f(x) = -\infty, \quad \lim_{x \rightarrow 3^+} f(x) = 2,$$

f is continuous from the right at 3

11–28 Find the limit.

11. $\lim_{x \rightarrow \infty} \frac{1-x}{2+5x}$
12. $\lim_{t \rightarrow \infty} 3^{-2t}$
13. $\lim_{x \rightarrow 1} e^{x^3-x}$
14. $\lim_{x \rightarrow 3} \frac{x^2-9}{x^2+2x-3}$
15. $\lim_{x \rightarrow -3} \frac{x^2-9}{x^2+2x-3}$
16. $\lim_{x \rightarrow 1^+} \frac{x^2-9}{x^2+2x-3}$
17. $\lim_{h \rightarrow 0} \frac{(h-1)^3+1}{h}$
18. $\lim_{t \rightarrow 2} \frac{t^2-4}{t^3-8}$
19. $\lim_{r \rightarrow 9} \frac{\sqrt{r}}{(r-9)^4}$
20. $\lim_{v \rightarrow 4^+} \frac{4-v}{|4-v|}$
21. $\lim_{u \rightarrow 1} \frac{u^4-1}{u^3+5u^2-6u}$
22. $\lim_{x \rightarrow 3} \frac{\sqrt{x+6}-x}{x^3-3x^2}$
23. $\lim_{x \rightarrow \pi^-} \ln(\sin x)$
24. $\lim_{x \rightarrow -\infty} \frac{1-2x^2-x^4}{5+x-3x^4}$
25. $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2-9}}{2x-6}$
26. $\lim_{x \rightarrow \infty} e^{x-x^2}$
27. $\lim_{x \rightarrow \infty} (\sqrt{x^2+4x+1}-x)$
28. $\lim_{x \rightarrow 1} \left(\frac{1}{x-1} + \frac{1}{x^2-3x+2} \right)$

29. The **Michaelis-Menten equation** for the rate v of the enzymatic reaction of the concentration $[S]$ of a substrate S , in the case of the enzyme pepsin, is

$$v = \frac{0.50[S]}{3.0 \times 10^{-4} + [S]}$$

What is $\lim_{[S] \rightarrow \infty} v$? What is the meaning of the limit in this context?

30. Prove that $\lim_{x \rightarrow 0} x^2 \cos(1/x^2) = 0$.

31. Let

$$f(x) = \begin{cases} \sqrt{-x} & \text{if } x < 0 \\ 3-x & \text{if } 0 \leq x < 3 \\ (x-3)^2 & \text{if } x > 3 \end{cases}$$

(a) Evaluate each limit, if it exists.

- (i) $\lim_{x \rightarrow 0^+} f(x)$ (ii) $\lim_{x \rightarrow 0^-} f(x)$ (iii) $\lim_{x \rightarrow 0} f(x)$
- (iv) $\lim_{x \rightarrow 3^-} f(x)$ (v) $\lim_{x \rightarrow 3^+} f(x)$ (vi) $\lim_{x \rightarrow 3} f(x)$

(b) Where is f discontinuous?

(c) Sketch the graph of f .

32. Show that each function is continuous on its domain. State the domain.

(a) $g(x) = \frac{\sqrt{x^2-9}}{x^2-2}$ (b) $h(x) = xe^{\sin x}$

33–34 Use the Intermediate Value Theorem to show that there is a root of the equation in the given interval.

33. $2x^3 + x^2 + 2 = 0, \quad (-2, -1)$

34. $e^{-x^2} = x, \quad (0, 1)$

CASE STUDY 2a Hosts, Parasites, and Time-Travel



We are studying a model for the interaction between *Daphnia* and its parasite. Recall that there are two possible host genotypes (A and a) and two possible parasite genotypes (B and b). Parasites of type B can infect only hosts of type A, while parasites of type b can infect only hosts of type a. Here we will take equations that will be obtained in Case Studies 2b and 2d to explore the biological predictions that can be obtained from them.

In Case Study 2d we will derive the functions

(1a) $q(t) = \frac{1}{2} + M_q \cos(ct - \phi_q)$

(1b) $p(t) = \frac{1}{2} + M_p \cos(ct - \phi_p)$

where $q(t)$ is the predicted frequency of host genotype A at time t and $p(t)$ is the predicted frequency of parasite genotype B at time t . In these equations ϕ_q , ϕ_p , and c are positive constants, while M_q , M_p are positive constants that are strictly less than $\frac{1}{2}$.

1. Describe, in words, how the genotype frequencies of the host and parasite change over time. Provide an explanation, in biological terms, for these dynamics.
2. How do the constants M_q and M_p affect the pattern of genotype frequencies over time?
3. The constant c is determined by the consequences of infection, in terms of reproduction, for both the host and the parasite. A large difference in reproductive success between infected versus uninfected hosts makes c large. Likewise, a large difference in reproductive success between parasites that are unable to infect a host versus those that are able to infect a host also makes c large. How does c affect the pattern of genotype frequencies over time as predicted by Equations 1? Provide an explanation for this in biological terms.
4. The constants ϕ_q and ϕ_p are referred to as the *phase* of $q(t)$ and $p(t)$, respectively. How do these constants affect the pattern of genotype frequencies over time?
5. As you will see in Equation 3, the difference $\phi^* = \phi_p - \phi_q$ turns out to be important in the coevolution of the host and parasite. What does this difference represent mathematically? Explain why this quantity is a measure of the extent to which the frequency of the parasite genotype lags behind the frequency of the host genotype.

Equations 1 give the genotype frequencies as functions of time. In the *Daphnia*-parasite system described in Case Study 2 on page xlvii, these equations can also be interpreted as giving the dynamics of genotype frequencies as a function of depth in the sediment core shown in Figure 1.

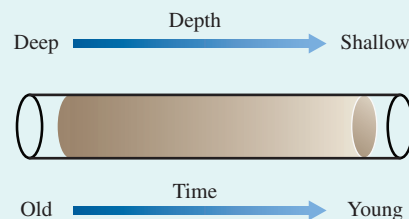


FIGURE 1

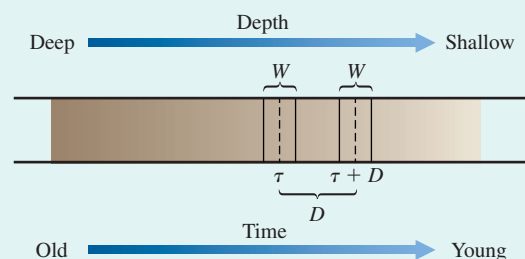


FIGURE 2

In the experiment described in Case Study 2, researchers chose a fixed depth τ and extracted a layer of sediment of width W centered around this depth (see Figure 2). The contents of this layer were mixed completely, and then hosts and parasites were extracted at random from the mixture. Researchers also took deeper and shallower layers (which represent the past and the future for hosts located in the layer at τ) and again completely mixed each layer. The center of these layers was a distance D from the center of the focal layer at τ , with $D < 0$ corresponding to a deeper layer and $D > 0$ a shallower layer (see Figure 2). The researchers then challenged hosts from the layer at τ with parasites from their past (that is, from the layer with $D < 0$), present (the layer at τ), and future (the layer with $D > 0$). For each challenge experiment the fraction of hosts becoming infected was measured.

In Case Study 2b we will show that, when a layer of sediment at location τ with width W is mixed completely, the frequency of type A hosts in this mixture is predicted to be

$$(2a) \quad q_{ave}(\tau) = \frac{1}{2} + M_q \cos(c\tau - \phi_q) \frac{2 \sin(\frac{1}{2}cW)}{cW}$$

Likewise, we will show that, when a layer of sediment at location τ with width W is mixed completely, the frequency of type B parasites in this mixture is predicted to be

$$(2b) \quad p_{ave}(\tau) = \frac{1}{2} + M_p \cos(c\tau - \phi_p) \frac{2 \sin(\frac{1}{2}cW)}{cW}$$

6. The functions (2a) and (2b) are similar to (1a) and (1b) except that the second terms are multiplied by the quantity $2 \sin(\frac{1}{2}cW)/cW$. Describe how the frequency of the genotypes within a mixed layer depends on the width W of this layer. In particular, what happens as the width of the layer becomes very small (that is, when $W \rightarrow 0$)? What happens as the width becomes very large (that is, $W \rightarrow \infty$)? Provide a biological interpretation for your answers.

In the experiment introduced in Case Study 2, hosts from depth τ were challenged with parasites from depth $\tau + D$. This was repeated for different depths τ , and the overall fraction of hosts infected was measured. In Case Study 2b we will show that the predicted fraction of hosts infected from such an experiment is

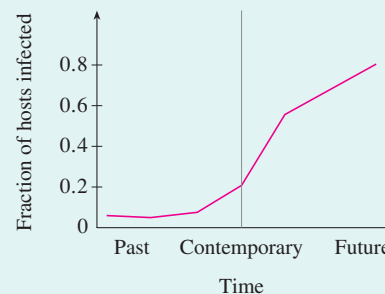
$$(3) \quad F(D) = \frac{1}{2} + M_p M_q \cos(cD - \phi^*) \frac{4 \sin^2(\frac{1}{2}cW)}{c^2 W^2}$$

where $\phi^* = \phi_p - \phi_q$ and $D < 0$ corresponds to parasites from a host's past and $D > 0$ to parasites from a host's future.

7. Sketch the graph of $F(D)$ when $\phi^* = 0$. Be as accurate as possible, showing where the maxima and minima occur as well as where the graph crosses the vertical axis. Construct similar sketches when ϕ^* is small and positive as well as when ϕ^* is small and negative. These plots depict the predicted fraction of infected hosts in the experiment as a function of the relative point in time from which the parasite was taken.
8. Suppose that cD is relatively small, meaning that the layers used in the challenge experiments are close to one another. Use your results from Problem 7 to explain how it is possible to obtain the experimental data like those shown in Figure 3.

FIGURE 3
Horizontal axis is the time from which the parasite was taken, relative to the host's point in time.

Source: Adapted from S. Gandon et al., "Host-Parasite Coevolution and Patterns of Adaptation across Time and Space," *Journal of Evolutionary Biology* 21 (2008): 1861–66.



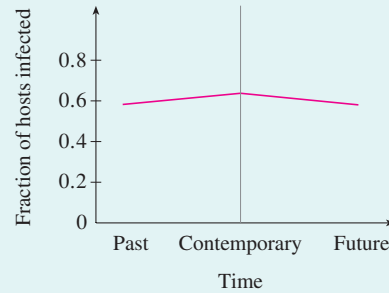
In particular, what is true about the value of ϕ^* in this case? Provide a biological interpretation of your answer.

9. Again suppose that cD is relatively small. Use your results from Problem 7 to explain how it is possible to obtain data like those shown in Figure 4. What is true about the value of ϕ^* in this case? Provide a biological interpretation of your answer.

FIGURE 4

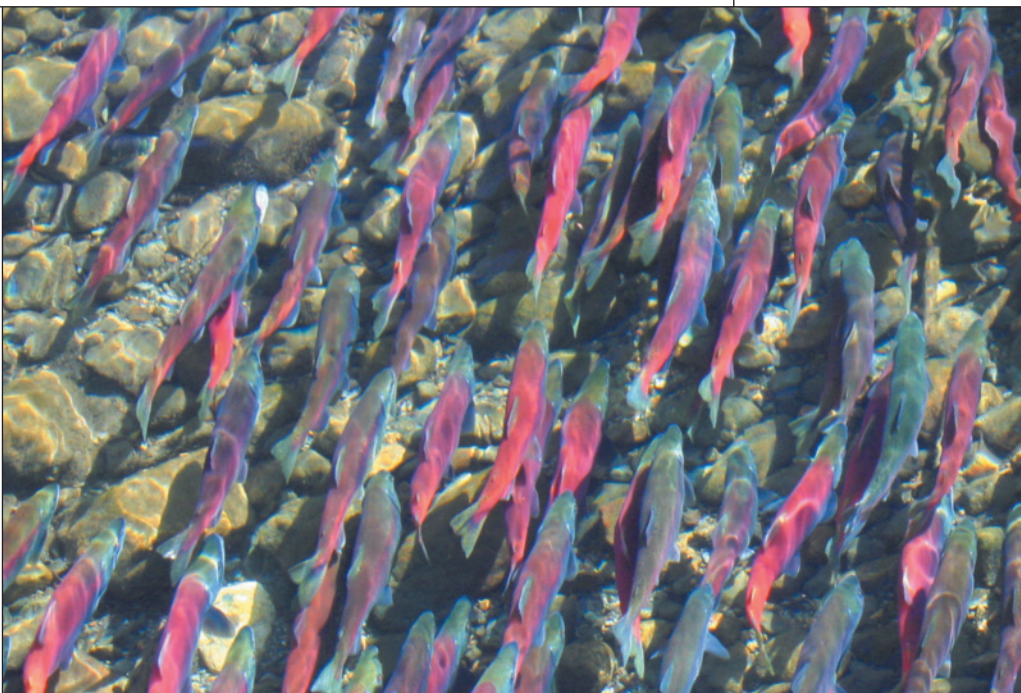
Horizontal axis is the time from which the parasite was taken, relative to the host's point in time.

Source: Adapted from S. Gandon et al., "Host-Parasite Coevolution and Patterns of Adaptation across Time and Space," *Journal of Evolutionary Biology* 21 (2008): 1861–66.



The maximum sustainable swimming speed S of salmon depends on the water temperature T . Exercise 42 in Section 3.1 asks you to analyze how S varies as T changes by estimating the derivative of S with respect to T .

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3.1 Derivatives and Rates of Change

3.2 The Derivative as a Function

3.3 Basic Differentiation Formulas

3.4 The Product and Quotient Rules

3.5 The Chain Rule

3.6 Exponential Growth and Decay

PROJECT: Controlling Red Blood Cell Loss During Surgery

3.7 Derivatives of the Logarithmic and Inverse Tangent Functions

3.8 Linear Approximations and Taylor Polynomials

PROJECT: Harvesting Renewable Resources

CASE STUDY 1b: Kill Curves and Antibiotic Effectiveness

IN THIS CHAPTER WE STUDY a special type of limit, called a derivative. Derivatives arise when we want to find a rate of growth, a velocity, the slope of a tangent line, or any instantaneous rate of change.

3.1 Derivatives and Rates of Change

The problem of finding a rate of change at a given instant and the problem of finding the tangent line to a curve at a given point involve finding the same type of limit, which we call a *derivative*.

■ Measuring the Rate of Increase of Blood Alcohol Concentration

Biomedical scientists have studied the chemical and physiological changes in the body that result from alcohol consumption. The reaction in the human body occurs in two stages: a fairly rapid process of absorption and a more gradual one of metabolism. To predict the effect of alcohol consumption, one needs to know the *rate* at which alcohol is absorbed and metabolized.

Medical researchers measured the blood alcohol concentration (BAC) of eight fasting adult male subjects after rapid consumption of 15 mL of ethanol (corresponding to one alcoholic drink).¹ The data they obtained were modeled by the concentration function

$$(1) \quad C(t) = 0.0225te^{-0.0467t}$$

where t is measured in minutes after consumption and C is measured in mg/mL. The graph of C is shown in Figure 1.

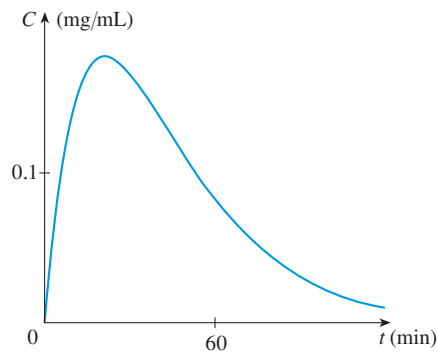


FIGURE 1

EXAMPLE 1 | Blood alcohol concentration How quickly is the BAC (given by Equation 1) increasing after 10 minutes?

SOLUTION We are asked to find the rate of change of C with respect to t when $t = 10$. The difficulty is that we are dealing with a single instant of time ($t = 10$ min) and so no time interval is involved. However, we can approximate the desired quantity

1. P. Wilkinson et al., "Pharmacokinetics of Ethanol after Oral Administration in the Fasting State," *Journal of Pharmacokinetics and Biopharmaceutics* 5 (1977): 207–24.

by calculating the *average* rate of change of C with respect to t in the time interval from $t = 10$ to $t = 11$:

$$\begin{aligned}\text{average rate of change} &= \frac{\text{change in } C}{\text{change in } t} = \frac{C(11) - C(10)}{11 - 10} \\ &\approx \frac{0.148073 - 0.141048}{1} = 0.007025 \text{ (mg/mL)/min}\end{aligned}$$

The following table shows the results of similar calculations of the average rates of change [in (mg/mL)/min] over successively smaller time periods.

Time interval	Average rate of change	Time interval	Average rate of change
$10 \leq t \leq 11$	0.00703	$9 \leq t \leq 10$	0.00804
$10 \leq t \leq 10.5$	0.00727	$9.5 \leq t \leq 10$	0.00777
$10 \leq t \leq 10.1$	0.00747	$9.9 \leq t \leq 10$	0.00757
$10 \leq t \leq 10.01$	0.00751	$9.99 \leq t \leq 10$	0.00752

It appears that as we shorten the time period, the average rate of change is becoming closer and closer to a number between 0.00752 and 0.00753 (mg/mL)/min. The *instantaneous rate of change* at $t = 10$ is defined to be the limiting value of these average rates of change over shorter and shorter time periods that start or end at $t = 10$. So we estimate that the BAC increased at a rate of about 0.0075 (mg/mL)/min. ■

We will return to the story about blood alcohol concentration in Example 6.

■ Tangent Lines

The word *tangent* is derived from the Latin word *tangens*, which means “touching.” Thus a tangent to a curve is a line that touches the curve. In other words, a tangent line should have the same direction as the curve at the point of contact. How can this idea be made precise?

For a circle we could simply follow Euclid and say that a tangent is a line that intersects the circle once and only once, as in Figure 2(a). For more complicated curves this definition is inadequate. Figure 2(b) shows two lines l and t passing through a point P on a curve C . The line l intersects C only once, but it certainly does not look like what we think of as a tangent. The line t , on the other hand, looks like a tangent but it intersects C twice.

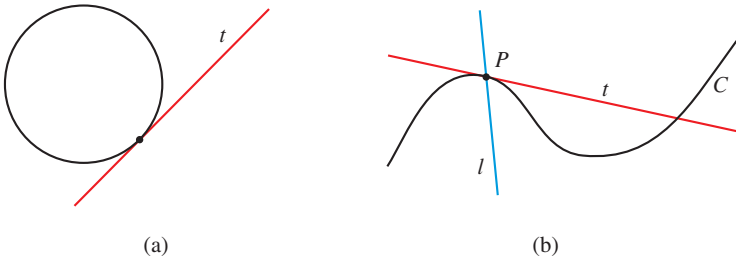


FIGURE 2

To be specific, let’s look at the problem of trying to find a tangent line t to the parabola $y = x^2$ in the following example.

EXAMPLE 2 | Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

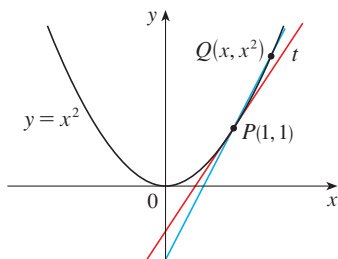


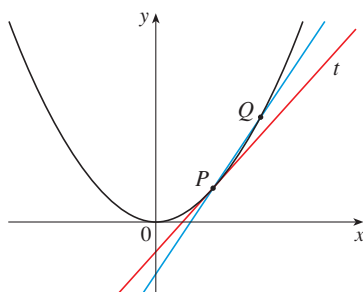
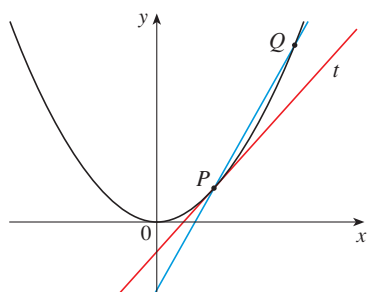
FIGURE 3

SOLUTION We will be able to find an equation of the tangent line t as soon as we know its slope m . The difficulty is that we know only one point, P , on t , whereas we need two points to compute the slope. But observe that we can compute an approximation to m by choosing a nearby point $Q(x, x^2)$ on the parabola (as in Figure 3) and computing the slope m_{PQ} of the secant line PQ . [A **secant line**, from the Latin word *secans*, meaning cutting, is a line that cuts (intersects) a curve more than once.]

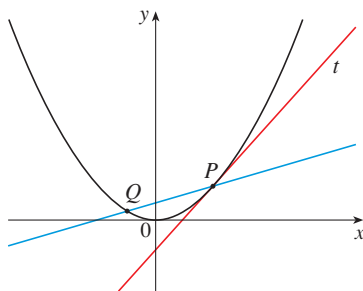
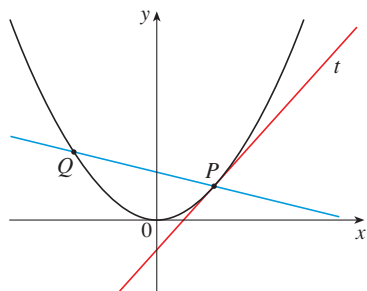
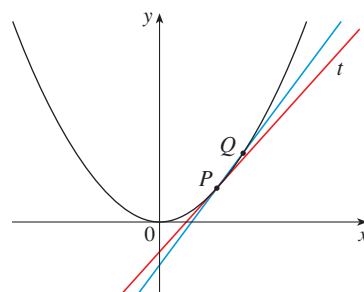
We choose $x \neq 1$ so that $Q \neq P$. Then

$$m_{PQ} = \frac{x^2 - 1}{x - 1}$$

What happens as x approaches 1? From Figure 4 we see that Q approaches P along the parabola and the secant lines PQ rotate about P and approach the tangent line t .



Q approaches P from the right



Q approaches P from the left

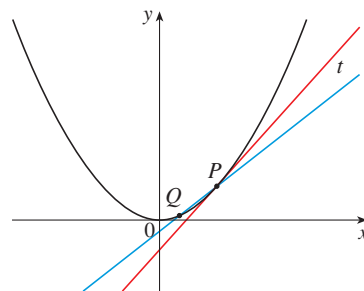


FIGURE 4

It appears that the slope m of the tangent line is the limit of the slopes of the secant lines as x approaches 1:

TEC In Visual 3.1A you can see how the process in Figure 4 works for additional functions.

$$\begin{aligned} m &= \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} \\ &= \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2 \end{aligned}$$

Point-slope form for a line through the point (x_1, y_1) with slope m :

$$y - y_1 = m(x - x_1)$$

Using the point-slope form of the equation of a line, we find that an equation of the tangent line at $(1, 1)$ is

$$y - 1 = 2(x - 1) \quad \text{or} \quad y = 2x - 1$$

TEC Visual 3.1B shows an animation of Figure 5.

We sometimes refer to the slope of the tangent line to a curve at a point as the **slope of the curve** at the point. The idea is that if we zoom in far enough toward the point, the curve looks almost like a straight line. Figure 5 illustrates this procedure for the curve $y = x^2$ in Example 2. The more we zoom in, the more the parabola looks like a line. In other words, the curve becomes almost indistinguishable from its tangent line.

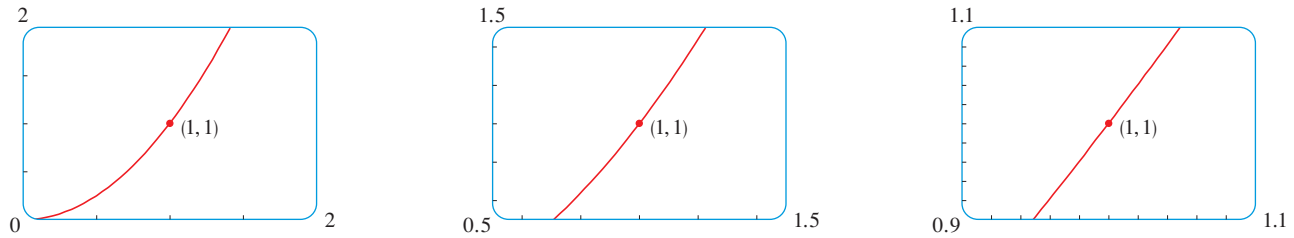


FIGURE 5 Zooming in toward the point $(1, 1)$ on the parabola $y = x^2$

In general, if a curve C has equation $y = f(x)$ and we want to find the tangent line to C at the point $P(a, f(a))$, then we consider a nearby point $Q(x, f(x))$, where $x \neq a$, and compute the slope of the secant line PQ :

$$m_{PQ} = \frac{f(x) - f(a)}{x - a}$$

Then we let Q approach P along the curve C by letting x approach a . If m_{PQ} approaches a number m , then we define the *tangent* t to be the line through P with slope m . (This amounts to saying that the tangent line is the limiting position of the secant line PQ as Q approaches P . See Figure 6.)

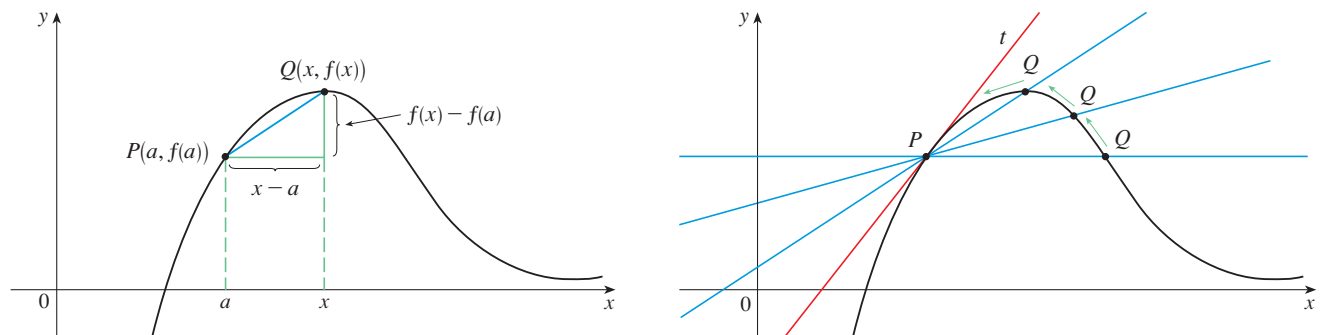


FIGURE 6

(2) Definition The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.

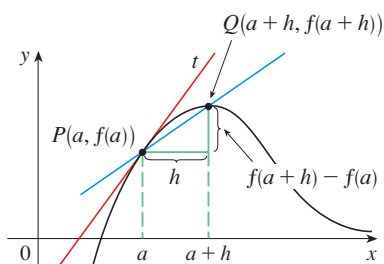


FIGURE 7

There is another expression for the slope of a tangent line that is sometimes easier to use. If $h = x - a$, then $x = a + h$ and so the slope of the secant line PQ is

$$m_{PQ} = \frac{f(a+h) - f(a)}{h}$$

(See Figure 7 where the case $h > 0$ is illustrated and Q is to the right of P . If it happened that $h < 0$, however, Q would be to the left of P .) Notice that as x approaches a , h approaches 0 (because $h = x - a$) and so the expression for the slope of the tangent line in Definition 2 becomes

(3)

$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

EXAMPLE 3 | Find an equation of the tangent line to the hyperbola $y = 3/x$ at the point $(3, 1)$.

SOLUTION Let $f(x) = 3/x$. Then, by Equation 3, the slope of the tangent at $(3, 1)$ is

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{\frac{3}{3+h} - 1}{h} = \lim_{h \rightarrow 0} \frac{\frac{3 - (3+h)}{3+h}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h(3+h)} = \lim_{h \rightarrow 0} -\frac{1}{3+h} = -\frac{1}{3} \end{aligned}$$

Therefore an equation of the tangent at the point $(3, 1)$ is

$$y - 1 = -\frac{1}{3}(x - 3)$$

which simplifies to $y = 2 - \frac{1}{3}x$ or $x + 3y - 6 = 0$

The hyperbola and its tangent are shown in Figure 8. ■

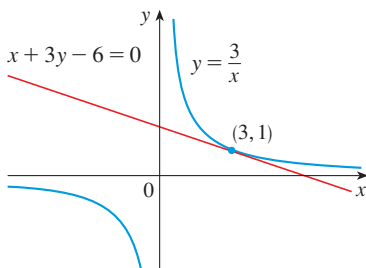


FIGURE 8

Derivatives

We have seen that the same type of limit arises in finding the slope of a tangent line (Equation 3) or a rate of change (Example 1). In fact, limits of the form

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

arise whenever we calculate a rate of change in any of the sciences, such as a rate of growth in biology or a rate of reaction in chemistry. Since this type of limit occurs so widely, it is given a special name and notation.

(4) Definition The **derivative of a function f at a number a** , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

$f'(a)$ is read “ f prime of a .”

If we write $x = a + h$, then we have $h = x - a$ and h approaches 0 if and only if x approaches a . Therefore an equivalent way of stating the definition of the derivative, as we saw in finding tangent lines, is

(5)

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

EXAMPLE 4 | Find the derivative of the function $f(x) = x^2 - 8x + 9$ at the number a .

SOLUTION From Definition 4 we have

$$\begin{aligned} f'(a) &= \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[(a + h)^2 - 8(a + h) + 9] - [a^2 - 8a + 9]}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^2 + 2ah + h^2 - 8a - 8h + 9 - a^2 + 8a - 9}{h} \\ &= \lim_{h \rightarrow 0} \frac{2ah + h^2 - 8h}{h} = \lim_{h \rightarrow 0} (2a + h - 8) \\ &= 2a - 8 \end{aligned}$$

We defined the tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ to be the line that passes through P and has slope m given by Equation 2 or 3. Since, by Definition 4, this is the same as the derivative $f'(a)$, we can now say the following.

The tangent line to $y = f(x)$ at $(a, f(a))$ is the line through $(a, f(a))$ whose slope is equal to $f'(a)$, the derivative of f at a .

If we use the point-slope form of the equation of a line, we can write an equation of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$:

$$y - f(a) = f'(a)(x - a)$$

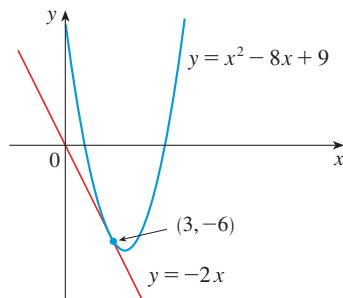


FIGURE 9

EXAMPLE 5 | Find an equation of the tangent line to the parabola $y = x^2 - 8x + 9$ at the point $(3, -6)$.

SOLUTION From Example 4 we know that the derivative of $f(x) = x^2 - 8x + 9$ at the number a is $f'(a) = 2a - 8$. Therefore the slope of the tangent line at $(3, -6)$ is $f'(3) = 2(3) - 8 = -2$. Thus an equation of the tangent line, shown in Figure 9, is

$$y - (-6) = (-2)(x - 3) \quad \text{or} \quad y = -2x$$

Rates of Change

We have already seen one example of a rate of change in Example 1. In general, suppose y is a quantity that depends on another quantity x . Thus y is a function of x and we write $y = f(x)$. If x changes from x_1 to x_2 , then the change in x (also called the **increment** of x) is

$$\Delta x = x_2 - x_1$$

and the corresponding change in y is

$$\Delta y = f(x_2) - f(x_1)$$

The difference quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

is called the **average rate of change of y with respect to x** over the interval $[x_1, x_2]$ and can be interpreted as the slope of the secant line PQ in Figure 10.

As we did in Example 1, we now consider the average rate of change over smaller and smaller intervals by letting x_2 approach x_1 and therefore letting Δx approach 0. The limit of these average rates of change is called the **(instantaneous) rate of change of y with respect to x at $x = x_1$** , which is interpreted as the slope of the tangent to the curve $y = f(x)$ at $P(x_1, f(x_1))$:

$$(6) \quad \text{instantaneous rate of change} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \rightarrow x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

We recognize this limit as being the derivative $f'(x_1)$.

We know that one interpretation of the derivative $f'(a)$ is as the slope of the tangent line to the curve $y = f(x)$ when $x = a$. We now have a second interpretation:

The derivative $f'(a)$ is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$.

The connection with the first interpretation is that if we sketch the curve $y = f(x)$, then the instantaneous rate of change is the slope of the tangent to this curve at the point where $x = a$. This means that when the derivative is large (and therefore the curve is steep, as at the point P in Figure 11), the y -values change rapidly. When the derivative is small, the curve is relatively flat (as at point Q) and the y -values change slowly.

EXAMPLE 6 | Blood alcohol concentration (continued) Draw the tangent line to the BAC curve in Example 1 at $t = 10$ and interpret its slope.

SOLUTION In Example 1 we estimated that the rate of increase of the blood alcohol concentration when $t = 10$ is about 0.0075 (mg/mL)/min. The equation of the curve (Equation 1) is

$$C(t) = 0.0225te^{-0.0467t}$$

which gives $C(10) \approx 0.14105$. So, using the point-slope equation of a line, we get that an approximate equation of the tangent line at $t = 10$ is

$$C - 0.14105 = 0.0075(t - 10)$$

or

$$C = 0.06605 + 0.0075t$$

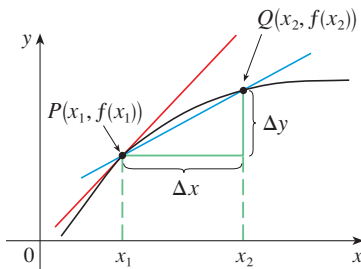


FIGURE 10

average rate of change = m_{PQ}
instantaneous rate of change
= slope of tangent at P

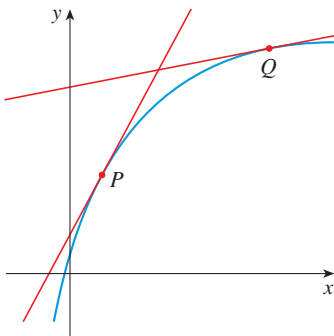


FIGURE 11

The y -values are changing rapidly at P and slowly at Q .

The concentration curve and its tangent line are graphed in Figure 12 and the slope of the tangent line is the rate of increase of BAC when $t = 10$.

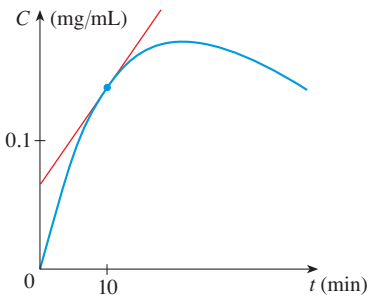


FIGURE 12

t	N
1	228
2	2,357
3	12,750
4	26,661
5	372,331
6	2,217,441

EXAMPLE 7 | **BB** **Malarial parasites** The table at the left, supplied by Andrew Read, shows experimental data involving malarial parasites. The time t is measured in days and N is the number of parasites per microliter of blood.

(a) Find the average rates of change of N with respect to t over the intervals $[1, 3]$, $[2, 3]$, $[3, 4]$, and $[3, 5]$.

(b) Interpret and estimate the value of the derivative $N'(3)$.

SOLUTION

(a) The average rate of change over $[1, 3]$ is

$$\frac{N(3) - N(1)}{3 - 1} = \frac{12,750 - 228}{2} = 6261 \text{ (parasites/}\mu\text{L)/day}$$

Similar calculations give the average rates of change in the following table:

Interval	Rate of change
$[1, 3]$	6,261
$[2, 3]$	10,393
$[3, 4]$	13,911
$[3, 5]$	179,791

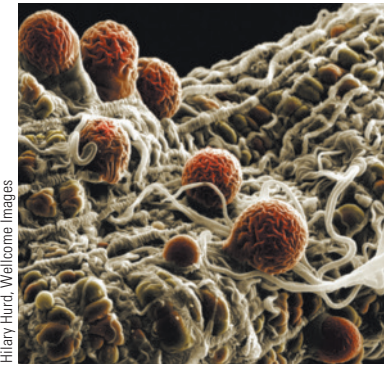
(b) The derivative $N'(3)$ means the rate of change of N with respect to t when $t = 3$ days. According to Equation 5,

$$N'(3) = \lim_{t \rightarrow 3} \frac{N(t) - N(3)}{t - 3}$$

The difference quotients in this expression (for various values of t) are just the rates of change in the table in part (a). So $N'(3)$ lies somewhere 10,393 and 13,911 (parasites/ μ L)/day. We estimate that the rate of increase of the parasite population on day 3 was approximately the average of these two numbers, namely

$$N'(3) \approx 12,152 \text{ (parasites/}\mu\text{L)/day}$$

A familiar example of a rate of change is **velocity**. In Example 2.3.1 we found the instantaneous velocity of a ball dropped from the CN Tower as the limit of average velocities over shorter and shorter time periods. More generally, if $s = f(t)$ is the position function of a particle that moves along a straight line, then $f'(a)$ is the rate of change



Although the function $N(t)$ is not a smooth function, it can be approximated by a smooth one as in Figure 2.5.12. In that sense it is meaningful to talk about the derivative $N'(3)$.

of the displacement s with respect to the time t . In other words, $f'(a)$ is the velocity of the particle at time $t = a$. The speed of the particle is the absolute value of the velocity, that is, $|f'(a)|$.

Let's revisit the example of the falling ball.

EXAMPLE 8 | Suppose that a ball is dropped from the upper observation deck of the CN Tower, 450 m above the ground.

- (a) What is the velocity of the ball after 5 seconds?
- (b) How fast is the ball traveling when it hits the ground?

SOLUTION We will need to find the velocity both when $t = 5$ and when the ball hits the ground, so it's efficient to start by finding the velocity at a general time t . Using the equation of motion $s = f(t) = 4.9t^2$, we have

$$\begin{aligned} v(t) = f'(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{4.9(t+h)^2 - 4.9t^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{4.9(t^2 + 2th + h^2 - t^2)}{h} = \lim_{h \rightarrow 0} \frac{4.9(2th + h^2)}{h} \\ &= \lim_{h \rightarrow 0} 4.9(2t + h) = 9.8t \end{aligned}$$

- (a) The velocity after 5 seconds is $v(5) = (9.8)(5) = 49$ m/s.
- (b) Since the observation deck is 450 m above the ground, the ball will hit the ground at the time t_1 when $s(t_1) = 450$, that is,

$$4.9t_1^2 = 450$$

This gives

$$t_1^2 = \frac{450}{4.9} \quad \text{and} \quad t_1 = \sqrt{\frac{450}{4.9}} \approx 9.6 \text{ s}$$

The velocity of the ball as it hits the ground is therefore

$$v(t_1) = 9.8t_1 = 9.8\sqrt{\frac{450}{4.9}} \approx 94 \text{ m/s}$$

EXAMPLE 9 | **HIV prevalence and incidence** In Example 2.4.7 we saw that the *prevalence* of a disease P is the number of cases as a function of time t . The *incidence* of the disease is the number of new infections per unit time over a specified time interval. We calculated the incidence of HIV in New York during the early 1980s over shorter and shorter time intervals. In the limiting case, as the interval shrinks to zero, we obtained the incidence of HIV at a point in time. Thus the incidence of HIV at a particular time can be viewed as the derivative of its prevalence function at that time. (Technically, this is true only if prevalence is changing solely due to the occurrence of new infections, as was the case for HIV in the early 1980s, since incidence is defined as the rate of generation of new infections. More generally, prevalence can change as a result of individuals dying or recovering from disease, as occurred later for HIV).

EXERCISES 3.1

1. A curve has equation $y = f(x)$.
 - (a) Write an expression for the slope of the secant line through the points $P(3, f(3))$ and $Q(x, f(x))$.
 - (b) Write an expression for the slope of the tangent line at P .
-  2. Graph the curve $y = e^x$ in the viewing rectangles $[-1, 1]$ by $[0, 2]$, $[-0.5, 0.5]$ by $[0.5, 1.5]$, and $[-0.1, 0.1]$ by $[0.9, 1.1]$. What do you notice about the curve as you zoom in toward the point $(0, 1)$?
3. (a) Find the slope of the tangent line to the parabola $y = 4x - x^2$ at the point $(1, 3)$
 - (i) using Definition 2 (ii) using Equation 3
 (b) Find an equation of the tangent line in part (a).
-  (c) Graph the parabola and the tangent line. As a check on your work, zoom in toward the point $(1, 3)$ until the parabola and the tangent line are indistinguishable.
4. (a) Find the slope of the tangent line to the curve $y = x - x^3$ at the point $(1, 0)$
 - (i) using Definition 2 (ii) using Equation 3
 (b) Find an equation of the tangent line in part (a).
-  (c) Graph the curve and the tangent line in successively smaller viewing rectangles centered at $(1, 0)$ until the curve and the line appear to coincide.

5–8 Find an equation of the tangent line to the curve at the given point.

5. $y = 4x - 3x^2$, $(2, -4)$

6. $y = x^3 - 3x + 1$, $(2, 3)$

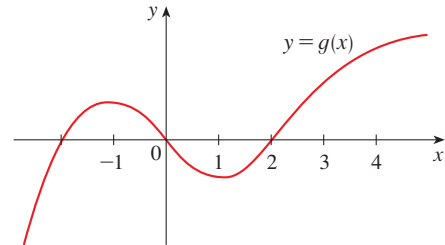
7. $y = \sqrt{x}$, $(1, 1)$

8. $y = \frac{2x + 1}{x + 2}$, $(1, 1)$

9. (a) Find the slope of the tangent to the curve $y = 3 + 4x^2 - 2x^3$ at the point where $x = a$.
- (b) Find equations of the tangent lines at the points $(1, 5)$ and $(2, 3)$.
-  (c) Graph the curve and both tangents on a common screen.
10. (a) Find the slope of the tangent to the curve $y = 1/\sqrt{x}$ at the point where $x = a$.
- (b) Find equations of the tangent lines at the points $(1, 1)$ and $(4, \frac{1}{2})$.
-  (c) Graph the curve and both tangents on a common screen.
11. For the function g whose graph is given, arrange the following numbers in increasing order and explain

your reasoning:

$$0 \quad g'(-2) \quad g'(0) \quad g'(2) \quad g'(4)$$



12. Find an equation of the tangent line to the graph of $y = g(x)$ at $x = 5$ if $g(5) = -3$ and $g'(5) = 4$.
13. If an equation of the tangent line to the curve $y = f(x)$ at the point where $a = 2$ is $y = 4x - 5$, find $f(2)$ and $f'(2)$.
14. If the tangent line to $y = f(x)$ at $(4, 3)$ passes through the point $(0, 2)$, find $f(4)$ and $f'(4)$.
15. Sketch the graph of a function f for which $f(0) = 0$, $f'(0) = 3$, $f'(1) = 0$, and $f'(2) = -1$.
16. Sketch the graph of a function g for which $g(0) = g(2) = g(4) = 0$, $g'(1) = g'(3) = 0$, $g'(0) = g'(4) = 1$, $g'(2) = -1$, $\lim_{x \rightarrow \infty} g(x) = \infty$, and $\lim_{x \rightarrow -\infty} g(x) = -\infty$.
17. If $f(x) = 3x^2 - x^3$, find $f'(1)$ and use it to find an equation of the tangent line to the curve $y = 3x^2 - x^3$ at the point $(1, 2)$.
18. If $g(x) = x^4 - 2$, find $g'(1)$ and use it to find an equation of the tangent line to the curve $y = x^4 - 2$ at the point $(1, -1)$.
19. (a) If $F(x) = 5x/(1 + x^2)$, find $F'(2)$ and use it to find an equation of the tangent line to the curve $y = 5x/(1 + x^2)$ at the point $(2, 2)$.
-  (b) Illustrate part (a) by graphing the curve and the tangent line on the same screen.
20. (a) If $G(x) = 4x^2 - x^3$, find $G'(a)$ and use it to find equations of the tangent lines to the curve $y = 4x^2 - x^3$ at the points $(2, 8)$ and $(3, 9)$.
-  (b) Illustrate part (a) by graphing the curve and the tangent lines on the same screen.

21–25 Find $f'(a)$.

21. $f(x) = 3x^2 - 4x + 1$

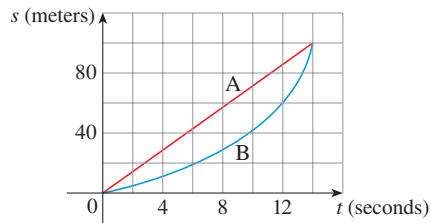
22. $f(t) = 2t^3 + t$

23. $f(t) = \frac{2t + 1}{t + 3}$

24. $f(x) = x^{-2}$

25. $f(x) = \sqrt{1 - 2x}$

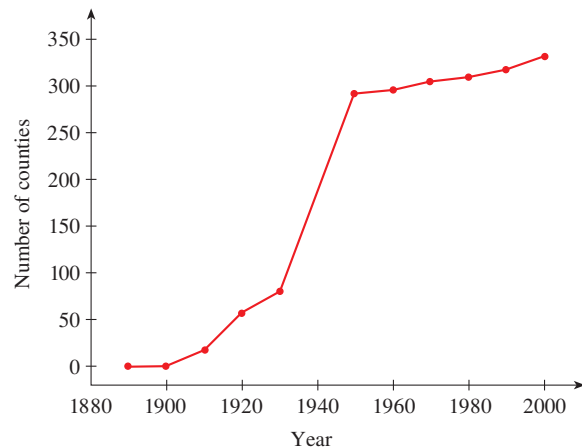
26. Shown are graphs of the position functions of two runners, A and B, who run a 100-meter race and finish in a tie.
- Describe and compare how the runners run the race.
 - At what time is the distance between the runners the greatest?
 - At what time do they have the same velocity?



27. If a ball is thrown into the air with a velocity of 40 ft/s, its height (in feet) after t seconds is given by $y = 40t - 16t^2$. Find the velocity when $t = 2$.
28. If a rock is thrown upward on the planet Mars with a velocity of 10 m/s, its height (in meters) after t seconds is given by $H = 10t - 1.86t^2$.
- Find the velocity of the rock after one second.
 - Find the velocity of the rock when $t = a$.
 - When will the rock hit the surface?
 - With what velocity will the rock hit the surface?
29. The displacement (in meters) of a particle moving in a straight line is given by the equation of motion $s = 1/t^2$, where t is measured in seconds. Find the velocity of the particle at times $t = a$, $t = 1$, $t = 2$, and $t = 3$.
30. **Invasive species** The Argentine ant is an invasive species in North America.



The graph shows the cumulative number of counties in the United States that have been invaded by this species over time.



- Estimate the average rate of invasion between 1890 and 1920, between 1920 and 1960, and between 1960 and 2000.
- Estimate the instantaneous rate of invasion in 1940.

31. **Population growth** The table gives the US midyear population, in millions, from 1990 to 2010.

t	1990	1995	2000	2005	2010
$P(t)$	249.6	266.3	282.2	295.8	308.3

- Find the average rate of population increase
 - from 1990 to 2000
 - from 1995 to 2000
 - from 2000 to 2005
 - from 2000 to 2010
- If $P(t)$ is the population at time t , estimate and interpret the value of the derivative $P'(2000)$.

32. **Viral load** The table shows values of the viral load $V(t)$ in HIV patient 303, measured in RNA copies/mL, t days after ABT-538 treatment was begun.

t	4	8	11	15	22
$V(t)$	53	18	9.4	5.2	3.6

- Find the average rate of change of V with respect to t over each time interval:
 - $[4, 11]$
 - $[8, 11]$
 - $[11, 15]$
 - $[11, 22]$
 What are the units?
- Estimate and interpret the value of the derivative $V'(11)$.

Source: Adapted from D. Ho et al., "Rapid Turnover of Plasma Virions and CD4 Lymphocytes in HIV-1 Infection," *Nature* 373 (1995): 123–26.

33. **Blood alcohol concentration** Researchers measured the average blood alcohol concentration $C(t)$ of eight men

starting one hour after consumption of 30 mL of ethanol (corresponding to two alcoholic drinks):

t (hours)	1.0	1.5	2.0	2.5	3.0
$C(t)$ (mg/mL)	0.33	0.24	0.18	0.12	0.07

- (a) Find the average rate of change of C with respect to t over each time interval:
- (i) $[1.0, 2.0]$
 - (ii) $[1.5, 2.0]$
 - (iii) $[2.0, 2.5]$
 - (iv) $[2.0, 3.0]$
- What are the units?
- (b) Estimate and interpret the value of the derivative $C'(2)$.

Source: Adapted from P. Wilkinson et al., "Pharmacokinetics of Ethanol after Oral Administration in the Fasting State," *Journal of Pharmacokinetics and Biopharmaceutics* 5 (1977): 207–24.

34. Let $D(t)$ be the US national debt at time t . The table gives approximate values of the function by providing end of year estimates, in billions of dollars, from 1990 to 2010. Interpret and estimate the value of $D'(2000)$.

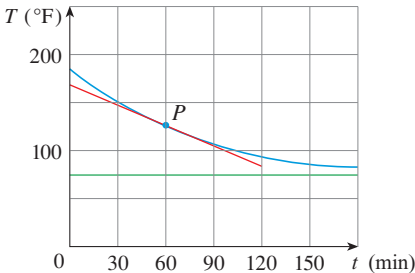
t	1990	1995	2000	2005	2010
$D(t)$	3233	4974	5662	8170	14,025

Source: US Dept. of the Treasury

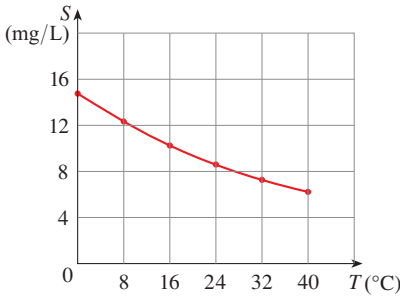
35. Let $T(t)$ be the temperature (in $^{\circ}\text{F}$) in Seattle t hours after midnight on May 7, 2012. The table shows values of this function recorded every two hours. What is the meaning of $T'(12)$? Estimate its value.

t	4	6	8	10	12	14	16
T	48	46	51	57	62	68	71

36. A roast turkey is taken from an oven when its temperature has reached 185°F and is placed on a table in a room where the temperature is 75°F . The graph shows how the temperature of the turkey decreases and eventually approaches room temperature. By measuring the slope of the tangent, estimate the rate of change of the temperature after an hour.



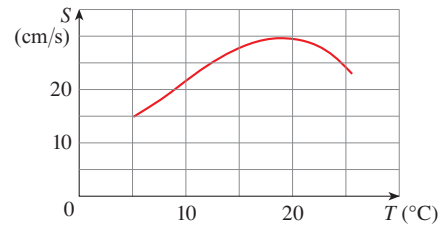
37. A warm can of soda is placed in a cold refrigerator. Sketch the graph of the temperature of the soda as a function of time. Is the initial rate of change of temperature greater or less than the rate of change after an hour?
38. **Bacteria population** The number of bacteria after t hours in a controlled laboratory experiment is $n = f(t)$.
- (a) What is the meaning of the derivative $f'(5)$? What are its units?
 - (b) Suppose there is an unlimited amount of space and nutrients for the bacteria. Which do you think is larger, $f'(5)$ or $f'(10)$? If the supply of nutrients is limited, would that affect your conclusion? Explain.
39. The cost of producing x ounces of gold from a new gold mine is $C = f(x)$ dollars.
- (a) What is the meaning of the derivative $f'(x)$? What are its units?
 - (b) What does the statement $f'(800) = 17$ mean?
 - (c) Do you think the values of $f'(x)$ will increase or decrease in the short term? What about the long term? Explain.
40. The quantity (in pounds) of a gourmet ground coffee that is sold by a coffee company at a price of p dollars per pound is $Q = f(p)$.
- (a) What is the meaning of the derivative $f'(8)$? What are its units?
 - (b) Is $f'(8)$ positive or negative? Explain.
41. **Oxygen solubility** The quantity of oxygen that can dissolve in water depends on the temperature of the water. (So thermal pollution influences the oxygen content of water.) The graph shows how oxygen solubility S varies as a function of the water temperature T .
- (a) What is the meaning of the derivative $S'(T)$? What are its units?
 - (b) Estimate the value of $S'(16)$ and interpret it.



Source: Adapted from C. Kupchella et al., *Environmental Science: Living Within the System of Nature*, 2d ed. (Boston: Allyn and Bacon, 1989).

42. Swimming speed of salmon The graph at the right shows the influence of the temperature T on the maximum sustainable swimming speed S of Coho salmon.

- (a) What is the meaning of the derivative $S'(T)$? What are its units?
 (b) Estimate the values of $S'(15)$ and $S'(25)$ and interpret them.



3.2 The Derivative as a Function

In the preceding section we considered the derivative of a function f at a fixed number a :

$$(1) \quad f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Here we change our point of view and let the number a vary. If we replace a in Equation 1 by a variable x , we obtain

$$(2) \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Given any number x for which this limit exists, we assign to x the number $f'(x)$. So we can regard f' as a new function, called the **derivative of f** and defined by Equation 2. We know that the value of f' at x , $f'(x)$, can be interpreted geometrically as the slope of the tangent line to the graph of f at the point $(x, f(x))$.

The function f' is called the derivative of f because it has been “derived” from f by the limiting operation in Equation 2. The domain of f' is the set $\{x \mid f'(x) \text{ exists}\}$ and may be smaller than the domain of f .

■ Graphing a Derivative from a Function's Graph

EXAMPLE 1 | The graph of a function f is given in Figure 1. Use it to sketch the graph of the derivative f' .

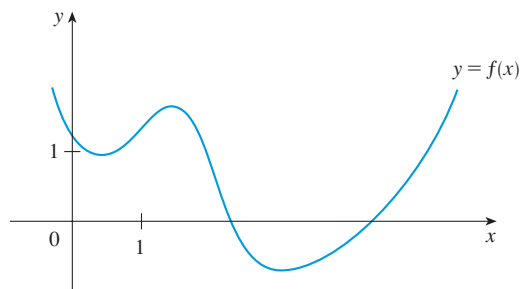


FIGURE 1

SOLUTION We can estimate the value of the derivative at any value of x by drawing the tangent at the point $(x, f(x))$ and estimating its slope. For instance, for $x = 5$ we draw the tangent at P in Figure 2(a) and estimate its slope to be about $\frac{3}{2}$, so $f'(5) \approx 1.5$. This allows us to plot the point $P'(5, 1.5)$ on the graph of f' directly beneath P . Repeating this procedure at several points, we get the graph shown in Figure 2(b). Notice that the tangents at A , B , and C are horizontal, so the derivative is 0 there and the graph of f' crosses the x -axis at the points A' , B' , and C' , directly beneath A , B , and C . Between A and B the tangents have positive slope, so $f'(x)$ is positive there. But between B and C the tangents have negative slope, so $f'(x)$ is negative there.

TEC Visual 3.2 shows an animation of Figure 2 for several functions.

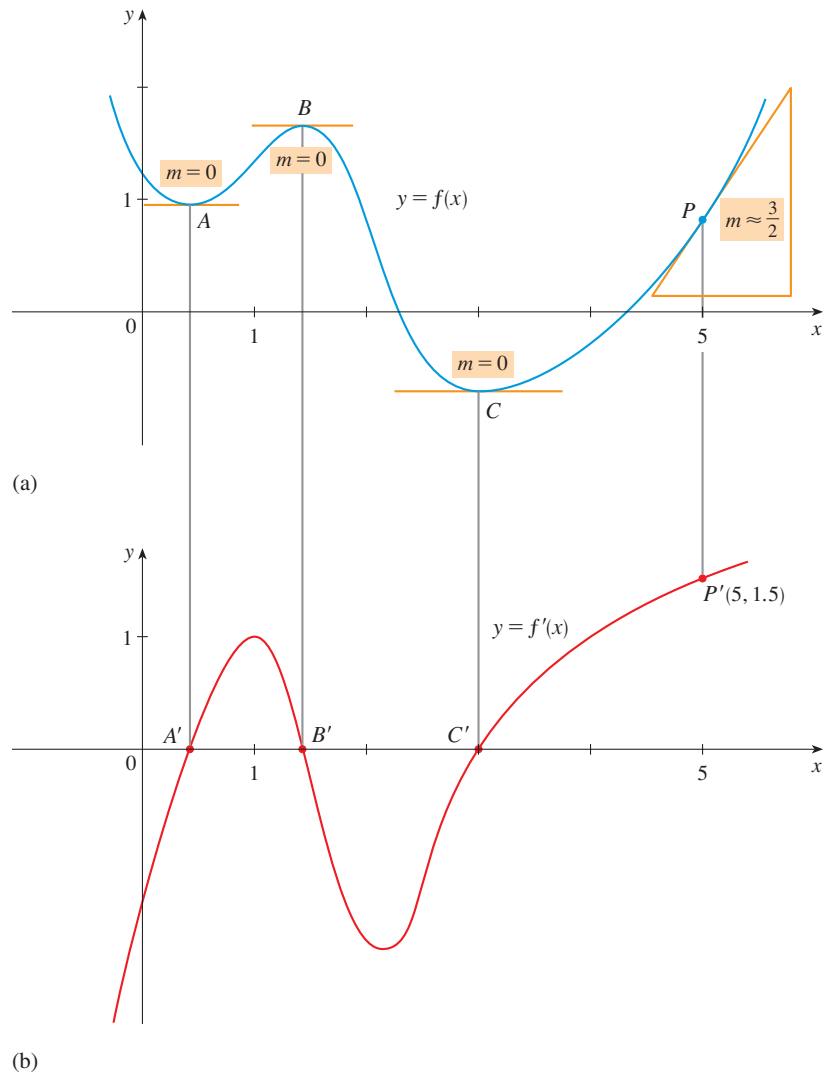


FIGURE 2

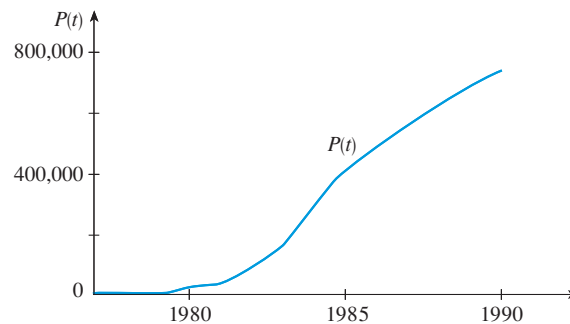
EXAMPLE 2 | HIV prevalence and incidence In epidemiology the *prevalence* of a disease, $P(t)$, is the number of individuals currently infected with the disease at time t . The *incidence* of a disease is the rate at which new individuals are contracting the disease. During the initial spread of HIV in the United States, prevalence

increased, changing only as a result of new infections. Consequently, $P'(t) \approx$ incidence from 1977–1990 (see Figure 3). Sketch a graph of the incidence of HIV over this time period.

FIGURE 3

HIV prevalence in the United States 1977–1990

Source: R. Song et al., “Estimation of HIV Incidence in the United States,” *Journal of the American Medical Association* 300 (2008): 520–29.

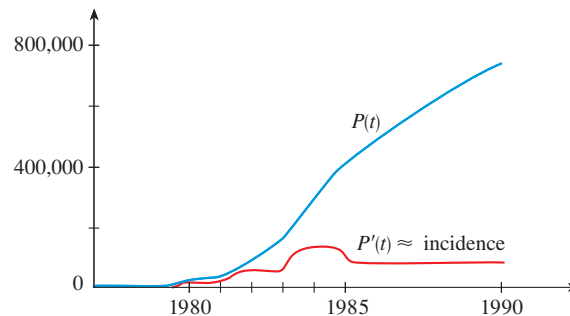


SOLUTION We take the incidence over time to be the function $P'(t)$. From the graph of $P(t)$ we see that the slope $P'(t)$ starts off near zero prior to 1980 (the rate of people becoming infected was small) and then increases. We see abrupt increases in the slope around 1981 and 1983, and an abrupt decrease around 1985, after which it remains relatively constant (see Figure 3). As a result, the incidence curve increases quickly in 1981 and again in 1983, and then decreases quickly in 1985 as shown in Figure 4. After 1985 it remains relatively flat, meaning that during this period a constant number of new infections per unit time occurred.

FIGURE 4

HIV prevalence and incidence in the United States 1977–1990

Source: R. Song et al., “Estimation of HIV Incidence in the United States,” *Journal of the American Medical Association* 300 (2008): 520–29.



■ Finding a Derivative from a Function's Formula

EXAMPLE 3

- If $f(x) = x^3 - x$, find a formula for $f'(x)$.
- Illustrate by comparing the graphs of f and f' .

SOLUTION

- When using Equation 2 to compute a derivative, we must remember that the variable is h and that x is temporarily regarded as a constant during the calculation of the limit.

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[(x+h)^3 - (x+h)] - [x^3 - x]}{h} \\
 &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x - h - x^3 + x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h} = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 - 1) = 3x^2 - 1
 \end{aligned}$$

(b) We use a graphing device to graph f and f' in Figure 5. Notice that $f'(x) = 0$ when f has horizontal tangents and $f'(x)$ is positive when the tangents have positive slope. So these graphs serve as a check on our work in part (a).

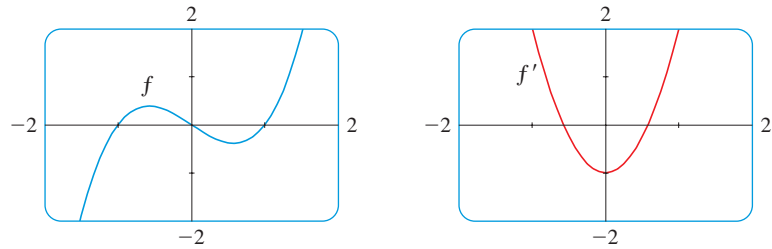


FIGURE 5

EXAMPLE 4 | If $f(x) = \sqrt{x}$, find the derivative of f . State the domain of f' .

SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \left(\frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \right) \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\ &= \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}} \end{aligned}$$

Here we rationalize the numerator.

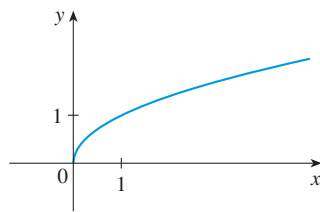
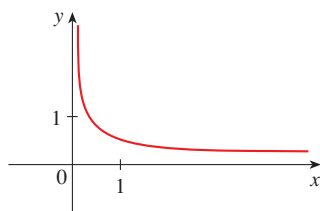
(a) $f(x) = \sqrt{x}$ (b) $f'(x) = \frac{1}{2\sqrt{x}}$

FIGURE 6

$$\frac{\frac{a}{b} - \frac{c}{d}}{e} = \frac{ad - bc}{bd} \cdot \frac{1}{e}$$

We see that $f'(x)$ exists if $x > 0$, so the domain of f' is $(0, \infty)$. This is smaller than the domain of f , which is $[0, \infty)$.

Let's check to see that the result of Example 4 is reasonable by looking at the graphs of f and f' in Figure 6. When x is close to 0, $\sqrt{x}$ is also close to 0, so $f'(x) = 1/(2\sqrt{x})$ is very large and this corresponds to the steep tangent lines near $(0, 0)$ in Figure 6(a) and the large values of $f'(x)$ just to the right of 0 in Figure 6(b). When x is large, $f'(x)$ is very small and this corresponds to the flatter tangent lines at the far right of the graph of f and the horizontal asymptote of the graph of f' .

EXAMPLE 5 | Find f' if $f(x) = \frac{1-x}{2+x}$.

SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1-(x+h)}{2+(x+h)} - \frac{1-x}{2+x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{(1-x-h)(2+x) - (1-x)(2+x+h)}{h(2+x+h)(2+x)} \\ &= \lim_{h \rightarrow 0} \frac{(2-x-2h-x^2-xh) - (2-x+h-x^2-xh)}{h(2+x+h)(2+x)} \\ &= \lim_{h \rightarrow 0} \frac{-3h}{h(2+x+h)(2+x)} = \lim_{h \rightarrow 0} \frac{-3}{(2+x+h)(2+x)} = -\frac{3}{(2+x)^2} \end{aligned}$$

■ Differentiability

If we use the traditional notation $y = f(x)$ to indicate that the independent variable is x and the dependent variable is y , then some common alternative notations for the derivative are as follows:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x)$$

Leibniz

Gottfried Wilhelm Leibniz was born in Leipzig in 1646 and studied law, theology, philosophy, and mathematics at the university there, graduating with a bachelor's degree at age 17. After earning his doctorate in law at age 20, Leibniz entered the diplomatic service and spent most of his life traveling to the capitals of Europe on political missions. In particular, he worked to avert a French military threat against Germany and attempted to reconcile the Catholic and Protestant churches.

His serious study of mathematics did not begin until 1672 while he was on a diplomatic mission in Paris. There he built a calculating machine and met scientists, like Huygens, who directed his attention to the latest developments in mathematics and science. Leibniz sought to develop a symbolic logic and system of notation that would simplify logical reasoning. In particular, the version of calculus that he published in 1684 established the notation and the rules for finding derivatives that we use today.

Unfortunately, a dreadful priority dispute arose in the 1690s between the followers of Newton and those of Leibniz as to who had invented calculus first. Leibniz was even accused of plagiarism by members of the Royal Society in England. The truth is that each man invented calculus independently. Newton arrived at his version of calculus first but, because of his fear of controversy, did not publish it immediately. So Leibniz's 1684 account of calculus was the first to be published.

The symbols D and d/dx are called **differentiation operators** because they indicate the operation of **differentiation**, which is the process of calculating a derivative.

The symbol dy/dx , which was introduced by Leibniz, should not be regarded as a ratio (for the time being); it is simply a synonym for $f'(x)$. Nonetheless, it is a very useful and suggestive notation, especially when used in conjunction with increment notation. Referring to Equation 3.1.6, we can rewrite the definition of derivative in Leibniz notation in the form

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

If we want to indicate the value of a derivative dy/dx in Leibniz notation at a specific number a , we use the notation

$$\left. \frac{dy}{dx} \right|_{x=a} \quad \text{or} \quad \left. \frac{dy}{dx} \right|_{x=a}$$

which is a synonym for $f'(a)$.

(3) Definition A function f is **differentiable at a** if $f'(a)$ exists. It is **differentiable on an open interval (a, b)** [or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

EXAMPLE 6 | Where is the function $f(x) = |x|$ differentiable?

SOLUTION If $x > 0$, then $|x| = x$ and we can choose h small enough that $x + h > 0$ and hence $|x + h| = x + h$. Therefore, for $x > 0$, we have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{|x + h| - |x|}{h} = \lim_{h \rightarrow 0} \frac{(x + h) - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} 1 = 1 \end{aligned}$$

and so f is differentiable for any $x > 0$.

Similarly, for $x < 0$ we have $|x| = -x$ and h can be chosen small enough that $x + h < 0$ and so $|x + h| = -(x + h)$. Therefore, for $x < 0$,

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{|x + h| - |x|}{h} = \lim_{h \rightarrow 0} \frac{-(x + h) - (-x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h} = \lim_{h \rightarrow 0} (-1) = -1 \end{aligned}$$

and so f is differentiable for any $x < 0$.

For $x = 0$ we have to investigate

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{|0+h| - |0|}{h} \quad (\text{if it exists}) \end{aligned}$$

Let's compute the left and right limits separately:

$$\lim_{h \rightarrow 0^+} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} 1 = 1$$

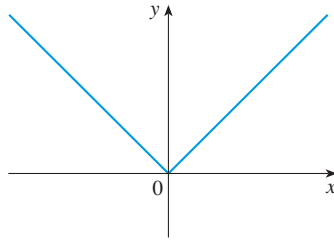
$$\text{and } \lim_{h \rightarrow 0^-} \frac{|0+h| - |0|}{h} = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = \lim_{h \rightarrow 0^-} (-1) = -1$$

Since these limits are different, $f'(0)$ does not exist. Thus f is differentiable at all x except 0.

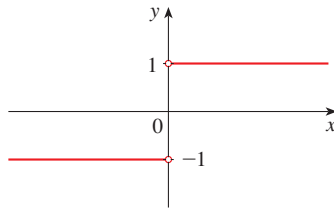
A formula for f' is given by

$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

and its graph is shown in Figure 7(b). The fact that $f'(0)$ does not exist is reflected geometrically in the fact that the curve $y = |x|$ does not have a tangent line at $(0, 0)$. [See Figure 7(a).]



(a) $y = f(x) = |x|$



(b) $y = f'(x)$

FIGURE 7

Both continuity and differentiability are desirable properties for a function to have. The following theorem shows how these properties are related.

(4) Theorem If f is differentiable at a , then f is continuous at a .

PROOF To prove that f is continuous at a , we have to show that $\lim_{x \rightarrow a} f(x) = f(a)$. We do this by first showing that the difference $f(x) - f(a)$ approaches 0.

The given information is that f is differentiable at a , that is,

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists (see Equation 3.1.5). To connect the given and the unknown, we divide and multiply $f(x) - f(a)$ by $x - a$ (which we can do when $x \neq a$):

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} (x - a)$$

Thus, using the Product Law and (3.1.5), we can write

$$\begin{aligned} \lim_{x \rightarrow a} [f(x) - f(a)] &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} (x - a) \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) \\ &= f'(a) \cdot 0 = 0 \end{aligned}$$

To use what we have just proved, we start with $f(x)$ and add and subtract $f(a)$:

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} [f(a) + (f(x) - f(a))] \\ &= \lim_{x \rightarrow a} f(a) + \lim_{x \rightarrow a} [f(x) - f(a)] \\ &= f(a) + 0 = f(a)\end{aligned}$$

Therefore f is continuous at a . ■

❗ **NOTE** The converse of Theorem 4 is false; that is, there are functions that are continuous but not differentiable. For instance, the function $f(x) = |x|$ is continuous at 0 because

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} |x| = 0 = f(0)$$

(See Example 2.4.8.) But in Example 6 we showed that f is not differentiable at 0.

We saw that the function $y = |x|$ in Example 6 is not differentiable at 0 and Figure 7(a) shows that its graph changes direction abruptly when $x = 0$. In general, if the graph of a function f has a “corner” or “kink” in it, then the graph of f has no tangent at this point and f is not differentiable there. [In trying to compute $f'(a)$, we find that the left and right limits are different.]

Theorem 4 gives another way for a function not to have a derivative. It says that if f is not continuous at a , then f is not differentiable at a . So at any discontinuity (for instance, a jump discontinuity) f fails to be differentiable.

A third possibility is that the curve has a **vertical tangent line** when $x = a$; that is, f is continuous at a and

$$\lim_{x \rightarrow a} |f'(x)| = \infty$$

This means that the tangent lines become steeper and steeper as $x \rightarrow a$. Figure 8 shows one way that this can happen; Figure 9(c) shows another. Figure 9 illustrates the three possibilities that we have discussed.

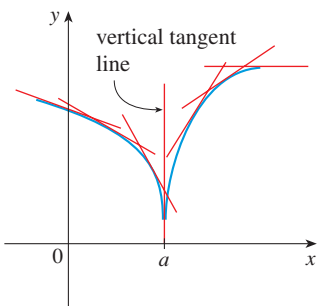


FIGURE 8

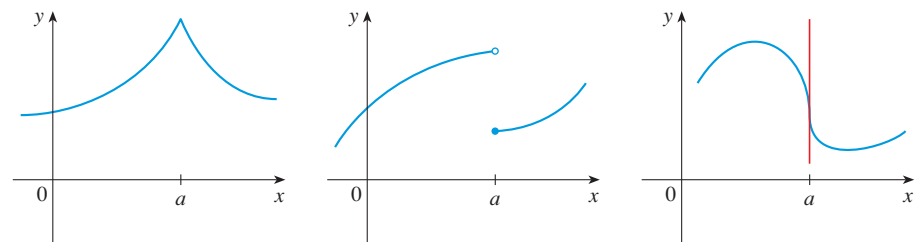


FIGURE 9

Three ways for f not to be differentiable at a

(a) A corner

(b) A discontinuity

(c) A vertical tangent

A graphing calculator or computer provides another way of looking at differentiability. If f is differentiable at a , then when we zoom in toward the point $(a, f(a))$ the graph straightens out and appears more and more like a line. (See Figure 10. We saw a specific example of this in Figure 3.1.5.) But no matter how much we zoom in toward a point like the ones in Figures 8 and 9(a), we can't eliminate the sharp point or corner (see Figure 11).

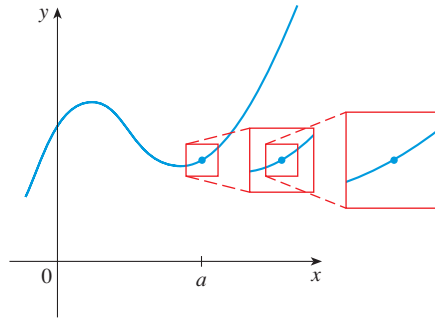


FIGURE 10
 f is differentiable at a .

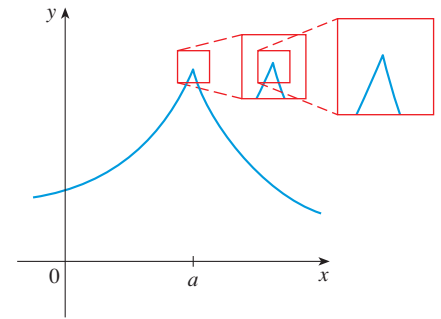


FIGURE 11
 f is not differentiable at a .

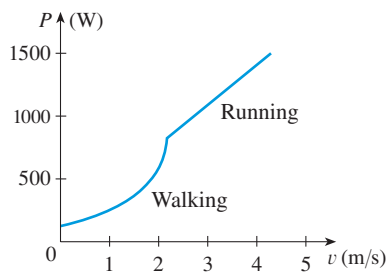


FIGURE 12

EXAMPLE 7 | **BB** Metabolic power in walking and running Figure 12 shows a graph of the metabolic power $P(v)$ consumed by humans who walk and then run at speed v .

- Is P a differentiable function of v ?
- Sketch the graph of $P'(v)$.

SOLUTION

- We see from the graph of P in Figure 12 that P is not differentiable at speed $v \approx 2.2$ m/s because the graph has a corner there. But P is differentiable at all other speeds.
- Using the method of Example 1, we measure slopes at a few points (as in Figure 13) and plot the resulting points in Figure 14.

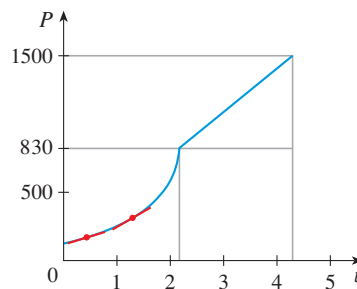


FIGURE 13

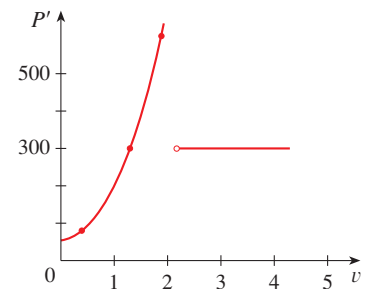


FIGURE 14

Notice, however, that unlike in Example 1, the axes have unequal scales. For instance, for $2.2 \leq v \leq 4.4$, the slope is approximately

$$\frac{1500 - 830}{4.4 - 2.2} = \frac{670}{2.2} \approx 305 \text{ W/(m/s)}$$

The discontinuity in the graph of P' at $v \approx 2.2$ reflects the fact that $P'(v)$ does not exist there.

Higher Derivatives

If f is a differentiable function, then its derivative f' is also a function, so f' may have a derivative of its own, denoted by $(f')' = f''$. This new function f'' is called the **second**

derivative of f because it is the derivative of the derivative of f . Using Leibniz notation, we write the second derivative of $y = f(x)$ as

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2}$$

EXAMPLE 8 | If $f(x) = x^3 - x$, find and interpret $f''(x)$.

SOLUTION In Example 3 we found that the first derivative is $f'(x) = 3x^2 - 1$. So the second derivative is

$$\begin{aligned} f''(x) &= (f')'(x) = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[3(x+h)^2 - 1] - [3x^2 - 1]}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 1 - 3x^2 + 1}{h} \\ &= \lim_{h \rightarrow 0} (6x + 3h) = 6x \end{aligned}$$

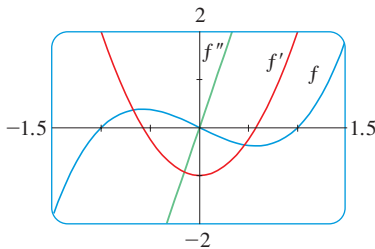


FIGURE 15

TEC In Module 3.2 you can see how changing the coefficients of a polynomial f affects the appearance of the graphs of f , f' , and f'' .

The graphs of f , f' , and f'' are shown in Figure 15.

We can interpret $f''(x)$ as the slope of the curve $y = f'(x)$ at the point $(x, f'(x))$. In other words, it is the rate of change of the slope of the original curve $y = f(x)$.

Notice from Figure 15 that $f''(x)$ is negative when $y = f'(x)$ has negative slope and positive when $y = f'(x)$ has positive slope. So the graphs serve as a check on our calculations. ■

In general, we can interpret a second derivative as a rate of change of a rate of change. The most familiar example of this is *acceleration*, which we define as follows.

If $s = s(t)$ is the position function of an object that moves in a straight line, we know that its first derivative represents the velocity $v(t)$ of the object as a function of time:

$$v(t) = s'(t) = \frac{ds}{dt}$$

The instantaneous rate of change of velocity with respect to time is called the **acceleration** $a(t)$ of the object. Thus the acceleration function is the derivative of the velocity function and is therefore the second derivative of the position function:

$$a(t) = v'(t) = s''(t)$$

or, in Leibniz notation,

$$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

The **third derivative** f''' is the derivative of the second derivative: $f''' = (f'')'$. So $f'''(x)$ can be interpreted as the slope of the curve $y = f''(x)$ or as the rate of change of $f''(x)$. If $y = f(x)$, then alternative notations for the third derivative are

$$y''' = f'''(x) = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{d^3y}{dx^3}$$

The process can be continued. The fourth derivative f'''' is usually denoted by $f^{(4)}$. In general, the n th derivative of f is denoted by $f^{(n)}$ and is obtained from f by differentiating n times. If $y = f(x)$, we write

$$y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n}$$

EXAMPLE 9 | If $f(x) = x^3 - x$, find $f'''(x)$ and $f^{(4)}(x)$.

SOLUTION In Example 8 we found that $f''(x) = 6x$. The graph of the second derivative has equation $y = 6x$ and so it is a straight line with slope 6. Since the derivative $f'''(x)$ is the slope of $f''(x)$, we have

$$f'''(x) = 6$$

for all values of x . So f''' is a constant function and its graph is a horizontal line. Therefore, for all values of x ,

$$f^{(4)}(x) = 0$$

■ What a Derivative Tells Us about a Function

Because $f'(x)$ represents the slope of the curve $y = f(x)$ at the point $(x, f(x))$, it tells us the direction in which the curve proceeds at each point. So it is reasonable to expect that information about $f'(x)$ will provide us with information about $f(x)$.

In particular, to see how the derivative of f can tell us where a function is increasing or decreasing, look at Figure 16. (Increasing functions and decreasing functions were defined in Section 1.1.) Between A and B and between C and D , the tangent lines have positive slope and so $f'(x) > 0$. Between B and C , the tangent lines have negative slope and so $f'(x) < 0$. Thus it appears that f increases when $f'(x)$ is positive and decreases when $f'(x)$ is negative.

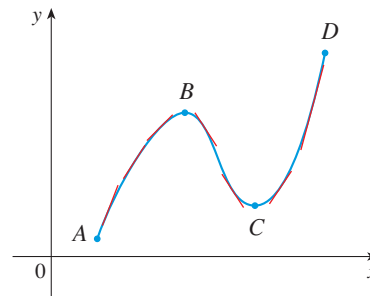


FIGURE 16

It turns out, as we will see in Chapter 4, that what we observed for the function graphed in Figure 16 is always true. We state the general result as follows.

If $f'(x) > 0$ on an interval, then f is increasing on that interval.

If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

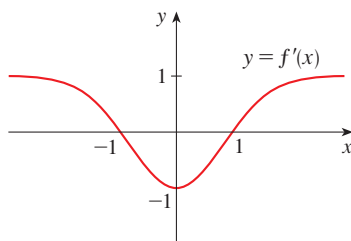


FIGURE 17

EXAMPLE 10

- If it is known that the graph of the derivative f' of a function is as shown in Figure 17, what can we say about f ?
- If it is known that $f(0) = 0$, sketch a possible graph of f .

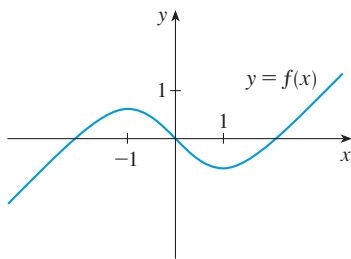


FIGURE 18

SOLUTION

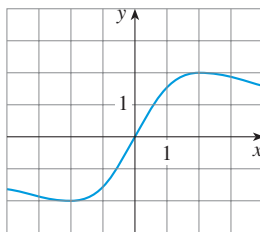
(a) We observe from Figure 17 that $f'(x)$ is negative when $-1 < x < 1$, so the original function f must be decreasing on the interval $(-1, 1)$. Similarly, $f'(x)$ is positive for $x < -1$ and for $x > 1$, so f is increasing on the intervals $(-\infty, -1)$ and $(1, \infty)$. Also note that, since $f'(-1) = 0$ and $f'(1) = 0$, the graph of f has horizontal tangents when $x = \pm 1$.

(b) We use the information from part (a), and the fact that the graph passes through the origin, to sketch a possible graph of f in Figure 18. Notice that $f'(0) = -1$, so we have drawn the curve $y = f(x)$ passing through the origin with a slope of -1 . Notice also that $f'(x) \rightarrow 1$ as $x \rightarrow \pm\infty$ (from Figure 17). So the slope of the curve $y = f(x)$ approaches 1 as x becomes large (positive or negative). That is why we have drawn the graph of f in Figure 18 progressively straighter as $x \rightarrow \pm\infty$.

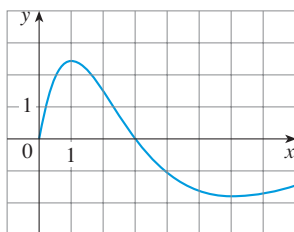
EXERCISES 3.2

1–2 Use the given graph to estimate the value of each derivative. Then sketch the graph of f' .

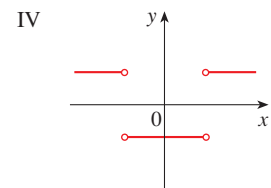
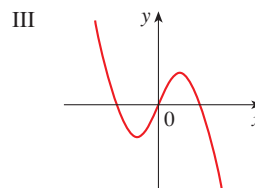
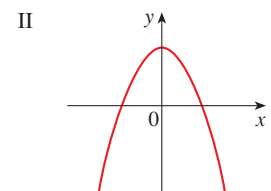
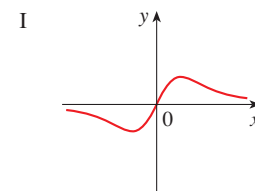
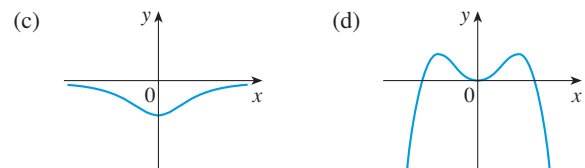
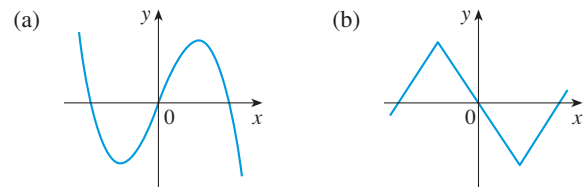
1. (a) $f'(-3)$ (b) $f'(-2)$
 (c) $f'(-1)$ (d) $f'(0)$
 (e) $f'(1)$ (f) $f'(2)$
 (g) $f'(3)$



2. (a) $f'(0)$ (b) $f'(1)$
 (c) $f'(2)$ (d) $f'(3)$
 (e) $f'(4)$ (f) $f'(5)$
 (g) $f'(6)$ (h) $f'(7)$

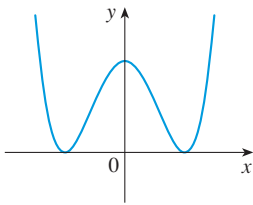


3. Match the graph of each function in (a)–(d) with the graph of its derivative in I–IV. Give reasons for your choices.

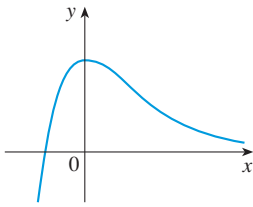


4–11 Trace or copy the graph of the given function f . (Assume that the axes have equal scales.) Then use the method of Example 1 to sketch the graph of f' below it.

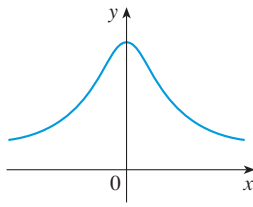
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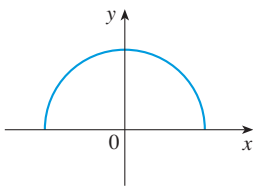
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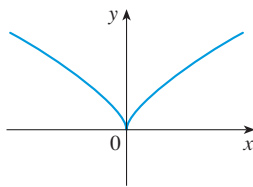
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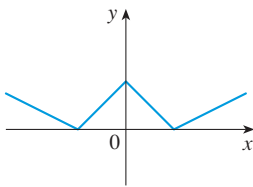
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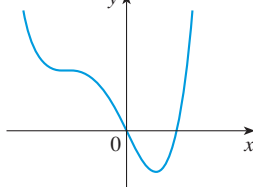
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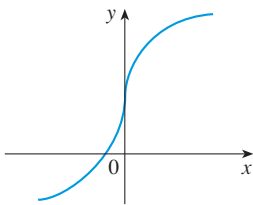
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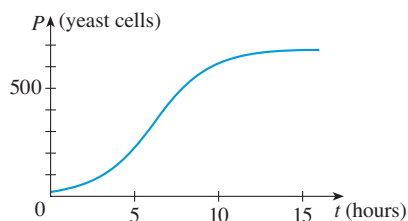
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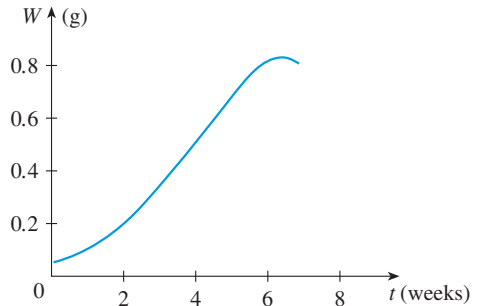


12. Yeast population Shown is the graph of the population function $P(t)$ for yeast cells in a laboratory culture. Use the method of Example 1 to graph the derivative $P'(t)$. What does the graph of P' tell us about the yeast population?



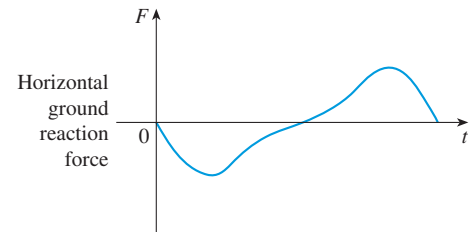
13. Tadpole weights The graph shows the average body weight W as a function of time for tadpoles raised in a density of 80 tadpoles/L.

- (a) What is the meaning of the derivative $W'(t)$?
(b) Sketch the graph of $W'(t)$.

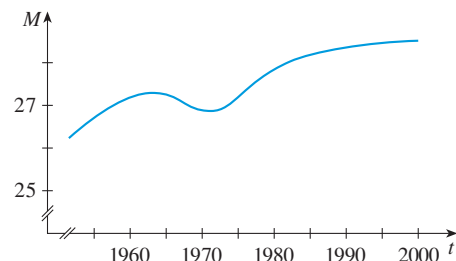


14. Ground reaction force in walking The graph shows the horizontal force $F(t)$ exerted by the ground on a person who is walking.

- (a) What is the meaning of the derivative $F'(t)$?
(b) Sketch the graph of $F'(t)$.



15. Marriage age The graph shows how the average age of first marriage of Japanese men varied in the last half of the 20th century. Sketch the graph of the derivative function $M'(t)$. During which years was the derivative negative?



16–18 Make a careful sketch of the graph of f and below it sketch the graph of f' in the same manner as in Exercises 4–11. Can you guess a formula for $f'(x)$ from its graph?

16. $f(x) = \sin x$

17. $f(x) = e^x$

18. $f(x) = \ln x$

-  19. Let $f(x) = x^2$.
- Estimate the values of $f'(0)$, $f'(\frac{1}{2})$, $f'(1)$, and $f'(2)$ by using a graphing device to zoom in on the graph of f .
 - Use symmetry to deduce the values of $f'(-\frac{1}{2})$, $f'(-1)$, and $f'(-2)$.
 - Use the results from parts (a) and (b) to guess a formula for $f'(x)$.
 - Use the definition of a derivative to prove that your guess in part (c) is correct.

-  20. Let $f(x) = x^3$.
- Estimate the values of $f'(0)$, $f'(\frac{1}{2})$, $f'(1)$, $f'(2)$, and $f'(3)$ by using a graphing device to zoom in on the graph of f .
 - Use symmetry to deduce the values of $f'(-\frac{1}{2})$, $f'(-1)$, $f'(-2)$, and $f'(-3)$.
 - Use the values from parts (a) and (b) to graph f' .
 - Guess a formula for $f'(x)$.
 - Use the definition of a derivative to prove that your guess in part (d) is correct.

21–31 Find the derivative of the function using the definition of a derivative. State the domain of the function and the domain of its derivative.

21. $f(x) = \frac{1}{2}x - \frac{1}{3}$
22. $f(x) = mx + b$
23. $f(t) = 5t - 9t^2$
24. $f(x) = 1.5x^2 - x + 3.7$
25. $f(x) = x^2 - 2x^3$
26. $f(x) = x + \sqrt{x}$
27. $g(x) = \sqrt{1 + 2x}$
28. $f(x) = \frac{x^2 - 1}{2x - 3}$
29. $G(t) = \frac{4t}{t + 1}$
30. $g(t) = \frac{1}{\sqrt{t}}$
31. $f(x) = x^4$

-  **32–34**
- Use the definition of the derivative to calculate f' .
 - Check to see that your answer is reasonable by comparing the graphs of f and f' .

32. $f(x) = x + 1/x$
33. $f(x) = x^4 + 2x$
34. $f(t) = t^2 - \sqrt{t}$

35. Malarial parasites An experiment measured the number of malarial parasites $N(t)$ per microliter of blood, where t is measured in days. The results of the experiment are shown in the table.

t	N	t	N
1	228	5	372,331
2	2357	6	2,217,441
3	12,750	7	6,748,400
4	26,661		

- What is the meaning of $N'(t)$? What are its units?
- Construct a table of estimated values for $N'(t)$.
[See Example 3.1.7 for $N'(3)$.]

36. Blood alcohol concentration Researchers measured the blood alcohol concentration $C(t)$ of eight adult male subjects after rapid consumption of 30 mL of ethanol (corresponding to two standard alcoholic drinks). The table shows the data they obtained by averaging the BAC (in mg/mL) of the eight men.

t (hours)	0.0	0.2	0.5	0.75	1.0	1.25
$C(t)$	0	0.25	0.41	0.40	0.33	0.29

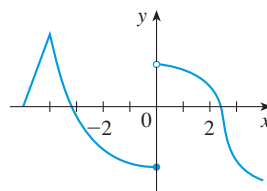
t (hours)	1.5	1.75	2.0	2.25	2.5	3.0
$C(t)$	0.24	0.22	0.18	0.15	0.12	0.07

- What is the meaning of $C'(t)$?
- Make a table of estimated values for $C'(t)$.

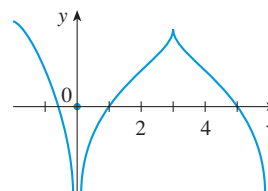
Source: Adapted from P. Wilkinson et al., "Pharmacokinetics of Ethanol after Oral Administration in the Fasting State," *Journal of Pharmacokinetics and Biopharmaceutics*, 5 (1977): 207–24.

37–40 The graph of f is given. State, with reasons, the numbers at which f is not differentiable.

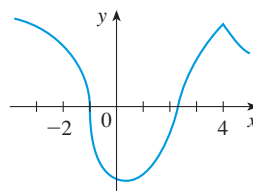
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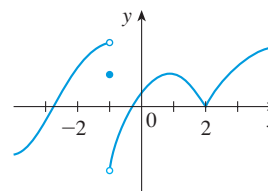
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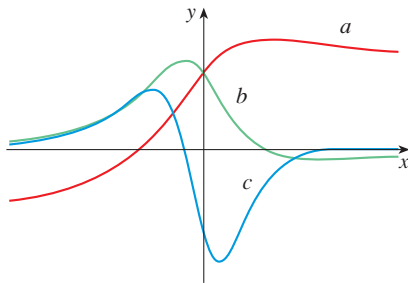


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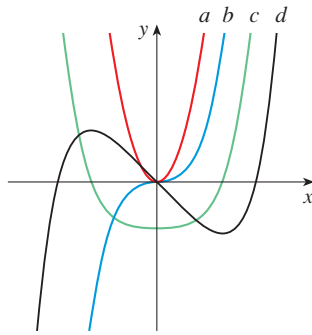


-  41. Graph the function $f(x) = x + \sqrt{|x|}$. Zoom in repeatedly, first toward the point $(-1, 0)$ and then toward the origin. What is different about the behavior of f in the vicinity of these two points? What do you conclude about the differentiability of f ?
-  42. Zoom in toward the points $(1, 0)$, $(0, 1)$, and $(-1, 0)$ on the graph of the function $g(x) = (x^2 - 1)^{2/3}$. What do you notice? Account for what you see in terms of the differentiability of g .

43. The figure shows the graphs of f , f' , and f'' . Identify each curve, and explain your choices.



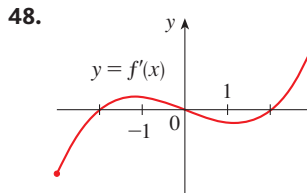
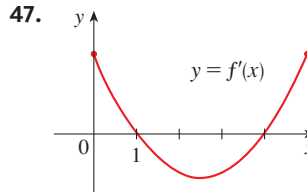
44. The figure shows graphs of f , f' , f'' , and f''' . Identify each curve, and explain your choices.



-  **45–46** Use the definition of a derivative to find $f'(x)$ and $f''(x)$. Then graph f , f' , and f'' on a common screen and check to see if your answers are reasonable.

45. $f(x) = 3x^2 + 2x + 1$ 46. $f(x) = x^3 - 3x$

- 47–48** The graph of the derivative f' of a function f is shown.
(a) On what intervals is f increasing?
(b) If it is known that $f(0) = 0$, sketch a possible graph of f .



49. Recall that a function f is called *even* if $f(-x) = f(x)$ for all x in its domain and *odd* if $f(-x) = -f(x)$ for all such x . Prove each of the following.
(a) The derivative of an even function is an odd function.
(b) The derivative of an odd function is an even function.
50. When you turn on a hot-water faucet, the temperature T of the water depends on how long the water has been running.
(a) Sketch a possible graph of T as a function of the time t that has elapsed since the faucet was turned on.
(b) Describe how the rate of change of T with respect to t varies as t increases.
(c) Sketch a graph of the derivative of T .

3.3 Basic Differentiation Formulas

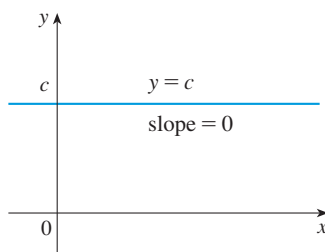


FIGURE 1
The graph of $f(x) = c$ is the line $y = c$, so $f'(x) = 0$.

If it were always necessary to compute derivatives directly from the definition, as we did in the preceding section, such computations would be tedious and the evaluation of some limits would require ingenuity. Fortunately, several rules have been developed for finding derivatives without having to use the definition directly. These formulas greatly simplify the task of differentiation.

In this section we learn how to differentiate constant functions, power functions, polynomials, exponential functions, and the sine and cosine functions.

Let's start with the simplest of all functions, the constant function $f(x) = c$. The graph of this function is the horizontal line $y = c$, which has slope 0, so we must have $f'(x) = 0$. (See Figure 1.) A formal proof, from the definition of a derivative, is also easy:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0$$

In Leibniz notation, we write this rule as follows.

Derivative of a Constant Function

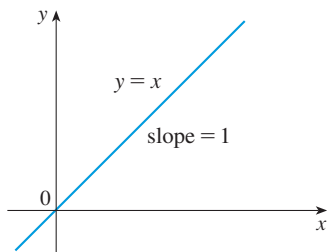
$$\frac{d}{dx}(c) = 0$$

Power Functions

We next look at the functions $f(x) = x^n$, where n is a positive integer. If $n = 1$, the graph of $f(x) = x$ is the line $y = x$, which has slope 1. (See Figure 2.) So

(1)

$$\frac{d}{dx}(x) = 1$$

**FIGURE 2**

The graph of $f(x) = x$ is the line $y = x$, so $f'(x) = 1$.

(You can also verify Equation 1 from the definition of a derivative.) We have already investigated the cases $n = 2$ and $n = 3$. In fact, in Section 3.2 (Exercises 19 and 20) we found that

(2)

$$\frac{d}{dx}(x^2) = 2x \quad \frac{d}{dx}(x^3) = 3x^2$$

For $n = 4$ we find the derivative of $f(x) = x^4$ as follows:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4 - x^4}{h} \\ &= \lim_{h \rightarrow 0} \frac{4x^3h + 6x^2h^2 + 4xh^3 + h^4}{h} \\ &= \lim_{h \rightarrow 0} (4x^3 + 6x^2h + 4xh^2 + h^3) = 4x^3 \end{aligned}$$

Thus

(3)

$$\frac{d}{dx}(x^4) = 4x^3$$

Comparing the equations in (1), (2), and (3), we see a pattern emerging. It seems to be a reasonable guess that, when n is a positive integer, $(d/dx)(x^n) = nx^{n-1}$. This turns out to be true.

The Power Rule If n is a positive integer, then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

PROOF If $f(x) = x^n$, then

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

The Binomial Theorem is given on Reference Page 1.

In finding the derivative of x^4 we had to expand $(x+h)^4$. Here we need to expand $(x+h)^n$ and we use the Binomial Theorem to do so:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\left[x^n + nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \cdots + nxh^{n-1} + h^n \right] - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \cdots + nxh^{n-1} + h^n}{h} \\ &= \lim_{h \rightarrow 0} \left[nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}h + \cdots + nxh^{n-2} + h^{n-1} \right] \\ &= nx^{n-1} \end{aligned}$$

because every term except the first has h as a factor and therefore approaches 0. ■

We illustrate the Power Rule using various notations in Example 1.

EXAMPLE 1

- (a) If $f(x) = x^6$, then $f'(x) = 6x^5$. (b) If $y = x^{1000}$, then $y' = 1000x^{999}$.
 (c) If $y = t^4$, then $\frac{dy}{dt} = 4t^3$. (d) $\frac{d}{dr}(r^3) = 3r^2$ ■

What about power functions with negative integer exponents? In Exercise 69 we ask you to verify from the definition of a derivative that

$$\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

We can rewrite this equation as

$$\frac{d}{dx} (x^{-1}) = (-1)x^{-2}$$

and so the Power Rule is true when $n = -1$. In fact, we will show in the next section [Exercise 3.4.63(c)] that it holds for all negative integers.

What if the exponent is a fraction? In Example 3.2.4 we found that

$$\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}$$

which can be written as

$$\frac{d}{dx} (x^{1/2}) = \frac{1}{2}x^{-1/2}$$

This shows that the Power Rule is true even when $n = \frac{1}{2}$. In fact, we will show in Section 3.7 that it is true for all real numbers n .

The Power Rule (General Version) If n is any real number, then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Figure 3 shows the function y in Example 2(b) and its derivative y' . Notice that y is not differentiable at 0 (y' is not defined there). Observe that y' is positive when y increases and is negative when y decreases.

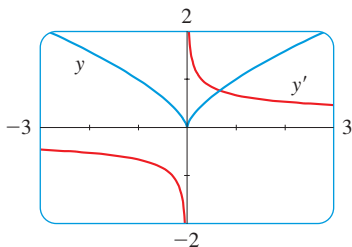


FIGURE 3
 $y = \sqrt[3]{x^2}$

EXAMPLE 2 | Differentiate

(a) $f(x) = \frac{1}{x^2}$ (b) $y = \sqrt[3]{x^2}$

SOLUTION In each case we rewrite the function as a power of x .

(a) Since $f(x) = x^{-2}$, we use the Power Rule with $n = -2$:

$$f'(x) = \frac{d}{dx}(x^{-2}) = -2x^{-2-1} = -2x^{-3} = -\frac{2}{x^3}$$

(b)
$$\frac{dy}{dx} = \frac{d}{dx}(\sqrt[3]{x^2}) = \frac{d}{dx}(x^{2/3}) = \frac{2}{3}x^{(2/3)-1} = \frac{2}{3}x^{-1/3}$$

The Power Rule enables us to find tangent lines without having to resort to the definition of a derivative. It also enables us to find *normal lines*. The **normal line** to a curve C at a point P is the line through P that is perpendicular to the tangent line at P . (In the study of optics, one needs to consider the angle between a light ray and the normal line to a lens.)

EXAMPLE 3 | Find equations of the tangent line and normal line to the curve $y = x\sqrt{x}$ at the point $(1, 1)$. Illustrate by graphing the curve and these lines.

SOLUTION The derivative of $f(x) = x\sqrt{x} = xx^{1/2} = x^{3/2}$ is

$$f'(x) = \frac{3}{2}x^{(3/2)-1} = \frac{3}{2}x^{1/2} = \frac{3}{2}\sqrt{x}$$

So the slope of the tangent line at $(1, 1)$ is $f'(1) = \frac{3}{2}$. Therefore an equation of the tangent line is

$$y - 1 = \frac{3}{2}(x - 1) \quad \text{or} \quad y = \frac{3}{2}x - \frac{1}{2}$$

The normal line is perpendicular to the tangent line, so its slope is the negative reciprocal of $\frac{3}{2}$, that is, $-\frac{2}{3}$. Thus an equation of the normal line is

$$y - 1 = -\frac{2}{3}(x - 1) \quad \text{or} \quad y = -\frac{2}{3}x + \frac{5}{3}$$

We graph the curve and its tangent line and normal line in Figure 4.

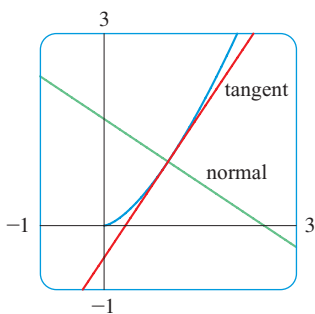
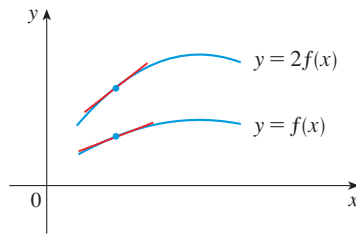


FIGURE 4
 $y = x\sqrt{x}$

■ New Derivatives from Old

When new functions are formed from old functions by addition, subtraction, or multiplication by a constant, their derivatives can be calculated in terms of derivatives of the old functions. In particular, the following formula says that *the derivative of a constant times a function is the constant times the derivative of the function*.

Geometric Interpretation of the Constant Multiple Rule

Multiplying by $c = 2$ stretches the graph vertically by a factor of 2. All the rises have been doubled but the runs stay the same. So the slopes are doubled, too.

The Constant Multiple Rule If c is a constant and f is a differentiable function, then

$$\frac{d}{dx} [cf(x)] = c \frac{d}{dx} f(x)$$

PROOF Let $g(x) = cf(x)$. Then

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\ &= \lim_{h \rightarrow 0} c \left[\frac{f(x+h) - f(x)}{h} \right] \\ &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (\text{by Limit Law 3}) \\ &= cf'(x) \end{aligned}$$

EXAMPLE 4

- (a) $\frac{d}{dx} (3x^4) = 3 \frac{d}{dx} (x^4) = 3(4x^3) = 12x^3$
- (b) $\frac{d}{dx} (-x) = \frac{d}{dx} [(-1)x] = (-1) \frac{d}{dx} (x) = -1(1) = -1$

EXAMPLE 5 | BB Anesthesiology As explained in Example 1.2.5, when ventilators are used during surgery the steady state concentration C of CO_2 in the lungs is $C = P/V$, where P is the rate of production of CO_2 by the body and V is the ventilation rate. If P is constant, find dC/dV and interpret it.

SOLUTION

Because P is constant, we can use the Constant Multiple Rule as follows:

$$\begin{aligned} \frac{dC}{dV} &= P \cdot \frac{d}{dV} \left(\frac{1}{V} \right) = P \cdot \frac{d}{dV} (V^{-1}) \\ &= P(-V^{-2}) = -\frac{P}{V^2} \end{aligned}$$

This is the rate of change of the concentration with respect to the ventilation rate. Notice the minus sign in the expression for dC/dV : the concentration decreases as the ventilation rate increases. Notice also that, because of the V^2 in the denominator, this rate of change is close to 0 when V is large.

The next rule tells us that *the derivative of a sum of functions is the sum of the derivatives*.

Using prime notation, we can write the Sum Rule as

$$(f + g)' = f' + g'$$

The Sum Rule If f and g are both differentiable, then

$$\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

PROOF Let $F(x) = f(x) + g(x)$. Then

$$\begin{aligned}
 F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} \\
 &= \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right] \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \quad (\text{by Limit Law 1}) \\
 &= f'(x) + g'(x)
 \end{aligned}$$

The Sum Rule can be extended to the sum of any number of functions. For instance, using this theorem twice, we get

$$(f + g + h)' = [(f + g) + h]' = (f + g)' + h' = f' + g' + h'$$

By writing $f - g$ as $f + (-1)g$ and applying the Sum Rule and the Constant Multiple Rule, we get the following formula.

The Difference Rule If f and g are both differentiable, then

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)$$

The Constant Multiple Rule, the Sum Rule, and the Difference Rule can be combined with the Power Rule to differentiate any polynomial, as the following examples demonstrate.

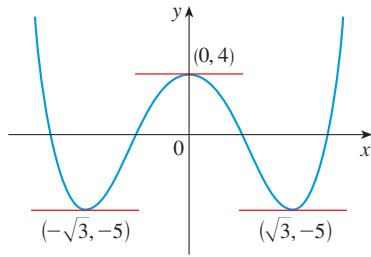
EXAMPLE 6

$$\begin{aligned}
 \frac{d}{dx}(x^8 + 12x^5 - 4x^4 + 10x^3 - 6x + 5) \\
 &= \frac{d}{dx}(x^8) + 12 \frac{d}{dx}(x^5) - 4 \frac{d}{dx}(x^4) + 10 \frac{d}{dx}(x^3) - 6 \frac{d}{dx}(x) + \frac{d}{dx}(5) \\
 &= 8x^7 + 12(5x^4) - 4(4x^3) + 10(3x^2) - 6(1) + 0 \\
 &= 8x^7 + 60x^4 - 16x^3 + 30x^2 - 6
 \end{aligned}$$

EXAMPLE 7 | Find the points on the curve $y = x^4 - 6x^2 + 4$ where the tangent line is horizontal.

SOLUTION Horizontal tangents occur where the derivative is zero. We have

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx}(x^4) - 6 \frac{d}{dx}(x^2) + \frac{d}{dx}(4) \\
 &= 4x^3 - 12x + 0 = 4x(x^2 - 3)
 \end{aligned}$$

**FIGURE 5**

The curve $y = x^4 - 6x^2 + 4$ and its horizontal tangents

Thus $dy/dx = 0$ if $x = 0$ or $x^2 - 3 = 0$, that is, $x = \pm\sqrt{3}$. So the given curve has horizontal tangents when $x = 0, \sqrt{3}$, and $-\sqrt{3}$. The corresponding points are $(0, 4)$, $(\sqrt{3}, -5)$, and $(-\sqrt{3}, -5)$. (See Figure 5.)

EXAMPLE 8 | The position function of a particle is $s = 2t^3 - 5t^2 + 3t + 4$, where s is measured in centimeters and t in seconds. Find the acceleration as a function of time. What is the acceleration after 2 seconds?

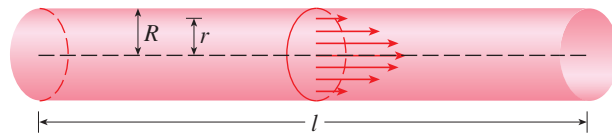
SOLUTION The velocity and acceleration are

$$v(t) = \frac{ds}{dt} = 6t^2 - 10t + 3$$

$$a(t) = \frac{dv}{dt} = 12t - 10$$

The acceleration after 2 s is $a(2) = 14 \text{ cm/s}^2$.

EXAMPLE 9 | **Blood flow** When we consider the flow of blood through a blood vessel, such as a vein or artery, we can model the shape of the blood vessel by a cylindrical tube with radius R and length l as illustrated in Figure 6.

**FIGURE 6**
Blood flow in an artery

Because of friction at the walls of the tube, the velocity v of the blood is greatest along the central axis of the tube and decreases as the distance r from the axis increases until v becomes 0 at the wall. The relationship between v and r is given by the **law of laminar flow** discovered by the French physician Jean-Louis-Marie Poiseuille in 1840. This law states that

$$(4) \quad v = \frac{P}{4\eta l} (R^2 - r^2)$$

For more detailed information, see W. Nichols and M. O'Rourke (eds.), *McDonald's Blood Flow in Arteries: Theoretical, Experimental, and Clinical Principles*, 5th ed. (New York, 2005).

where η is the viscosity of the blood and P is the pressure difference between the ends of the tube. If P and l are constant, then v is a function of r with domain $[0, R]$.

The average rate of change of the velocity as we move from $r = r_1$ outward to $r = r_2$ is given by

$$\frac{\Delta v}{\Delta r} = \frac{v(r_2) - v(r_1)}{r_2 - r_1}$$

and if we let $\Delta r \rightarrow 0$, we obtain the **velocity gradient**, that is, the instantaneous rate of change of velocity with respect to r :

$$\text{velocity gradient} = \lim_{\Delta r \rightarrow 0} \frac{\Delta v}{\Delta r} = \frac{dv}{dr}$$

Using Equation 4, we obtain

$$\frac{dv}{dr} = \frac{P}{4\eta l} (0 - 2r) = -\frac{Pr}{2\eta l}$$

For one of the smaller human arteries we can take $\eta = 0.027$, $R = 0.008$ cm, $l = 2$ cm, and $P = 4000$ dynes/cm², which gives

$$v = \frac{4000}{4(0.027)^2} (0.000064 - r^2)$$

$$\approx 1.85 \times 10^4 (6.4 \times 10^{-5} - r^2)$$

At $r = 0.002$ cm the blood is flowing at a speed of

$$v(0.002) \approx 1.85 \times 10^4 (64 \times 10^{-6} - 4 \times 10^{-6})$$

$$= 1.11 \text{ cm/s}$$

and the velocity gradient at that point is

$$\left. \frac{dv}{dr} \right|_{r=0.002} = -\frac{4000(0.002)}{2(0.027)^2} \approx -74 \text{ (cm/s)/cm}$$

To get a feeling for what this statement means, let's change our units from centimeters to micrometers ($1 \text{ cm} = 10,000 \text{ } \mu\text{m}$). Then the radius of the artery is $80 \text{ } \mu\text{m}$. The velocity at the central axis is $11,850 \text{ } \mu\text{m/s}$, which decreases to $11,110 \text{ } \mu\text{m/s}$ at a distance of $r = 20 \text{ } \mu\text{m}$. The fact that $dv/dr = -74 \text{ (}\mu\text{m/s)/}\mu\text{m}$ means that, when $r = 20 \text{ } \mu\text{m}$, the velocity is decreasing at a rate of about $74 \text{ } \mu\text{m/s}$ for each micrometer that we proceed away from the center. ■

■ Exponential Functions

Let's try to compute the derivative of the exponential function $f(x) = b^x$ using the definition of a derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{b^x b^h - b^x}{h} = \lim_{h \rightarrow 0} \frac{b^x (b^h - 1)}{h}$$

The factor b^x doesn't depend on h , so we can take it in front of the limit:

$$f'(x) = b^x \lim_{h \rightarrow 0} \frac{b^h - 1}{h}$$

Notice that the limit is the value of the derivative of f at 0, that is,

$$\lim_{h \rightarrow 0} \frac{b^h - 1}{h} = f'(0)$$

Therefore we have shown that if the exponential function $f(x) = b^x$ is differentiable at 0, then it is differentiable everywhere and

$$(5) \quad f'(x) = f'(0) b^x$$

This equation says that *the rate of change of any exponential function is proportional to the function itself*. (The slope is proportional to the height.)

h	$\frac{2^h - 1}{h}$	$\frac{3^h - 1}{h}$
0.1	0.7177	1.1612
0.01	0.6956	1.1047
0.001	0.6934	1.0992
0.0001	0.6932	1.0987

Numerical evidence for the existence of $f'(0)$ is given in the table at the left for the cases $b = 2$ and $b = 3$. (Values are stated correct to four decimal places.) It appears that the limits exist and

$$\text{for } b = 2, \quad f'(0) = \lim_{h \rightarrow 0} \frac{2^h - 1}{h} \approx 0.69$$

$$\text{for } b = 3, \quad f'(0) = \lim_{h \rightarrow 0} \frac{3^h - 1}{h} \approx 1.10$$

In fact, it can be proved that these limits exist and, correct to six decimal places, the values are

$$\left. \frac{d}{dx} (2^x) \right|_{x=0} \approx 0.693147 \quad \left. \frac{d}{dx} (3^x) \right|_{x=0} \approx 1.098612$$

Thus, from Equation 5, we have

$$(6) \quad \frac{d}{dx} (2^x) \approx (0.69)2^x \quad \frac{d}{dx} (3^x) \approx (1.10)3^x$$

Of all possible choices for the base b in Equation 5, the simplest differentiation formula occurs when $f'(0) = 1$. In view of the estimates of $f'(0)$ for $b = 2$ and $b = 3$, it seems reasonable that there is a number b between 2 and 3 for which $f'(0) = 1$. It is traditional to denote this value by the letter e . (In fact, that is how we introduced e in Section 1.4.) Thus we have the following definition.

In Exercise 1 we will see that e lies between 2.7 and 2.8. Later we will be able to show that, correct to five decimal places,

$$e \approx 2.71828$$

Definition of the Number e

$$e \text{ is the number such that } \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$$

Geometrically, this means that of all the possible exponential functions $y = b^x$, the function $f(x) = e^x$ is the one whose tangent line at $(0, 1)$ has a slope $f'(0)$ that is exactly 1. (See Figures 7 and 8.)

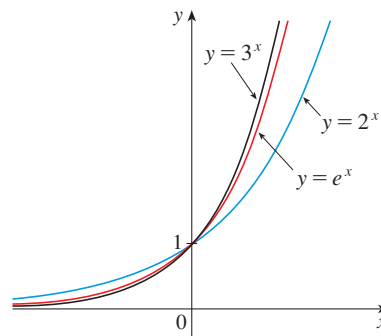


FIGURE 7

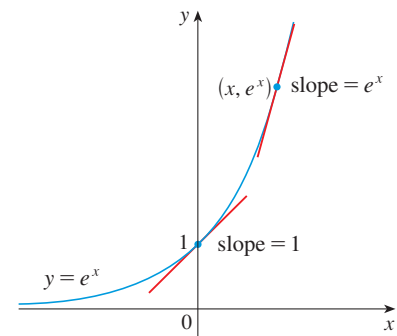


FIGURE 8

If we put $b = e$ and, therefore, $f'(0) = 1$ in Equation 5, it becomes the following important differentiation formula.

TEC Visual 3.3 uses the slope-a-scope to illustrate this formula.

Derivative of the Natural Exponential Function

$$\frac{d}{dx}(e^x) = e^x$$

Thus the exponential function $f(x) = e^x$ has the property that it is its own derivative. The geometrical significance of this fact is that the slope of a tangent line to the curve $y = e^x$ is equal to the y -coordinate of the point (see Figure 8).

EXAMPLE 10 | If $f(x) = e^x - x$, find f' and f'' . Compare the graphs of f and f' .

SOLUTION Using the Difference Rule, we have

$$f'(x) = \frac{d}{dx}(e^x - x) = \frac{d}{dx}(e^x) - \frac{d}{dx}(x) = e^x - 1$$

In Section 3.2 we defined the second derivative as the derivative of f' , so

$$f''(x) = \frac{d}{dx}(e^x - 1) = \frac{d}{dx}(e^x) - \frac{d}{dx}(1) = e^x$$

The function f and its derivative f' are graphed in Figure 9. Notice that f has a horizontal tangent when $x = 0$; this corresponds to the fact that $f'(0) = 0$. Notice also that, for $x > 0$, $f'(x)$ is positive and f is increasing. When $x < 0$, $f'(x)$ is negative and f is decreasing.

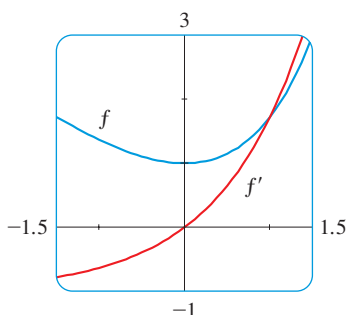


FIGURE 9

EXAMPLE 11 | At what point on the curve $y = e^x$ is the tangent line parallel to the line $y = 2x$?

SOLUTION Since $y = e^x$, we have $y' = e^x$. Let the x -coordinate of the point in question be a . Then the slope of the tangent line at that point is e^a . This tangent line will be parallel to the line $y = 2x$ if it has the same slope, that is, 2. Equating slopes, we get

$$e^a = 2 \quad \Rightarrow \quad a = \ln 2$$

Therefore the required point is $(a, e^a) = (\ln 2, 2)$. (See Figure 10.)

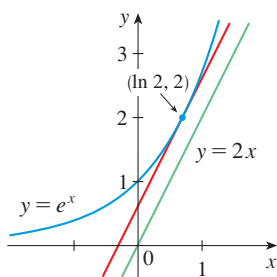


FIGURE 10

A review of the trigonometric functions is given in Appendix C.

Sine and Cosine Functions

If we sketch the graph of the function $f(x) = \sin x$ and use the interpretation of $f'(x)$ as the slope of the tangent to the sine curve in order to sketch the graph of f' (see Exercise 3.2.16), then it looks as if the graph of f' may be the same as the cosine curve (see Figure 11).

TEC Visual 3.3 shows an animation of Figure 11.

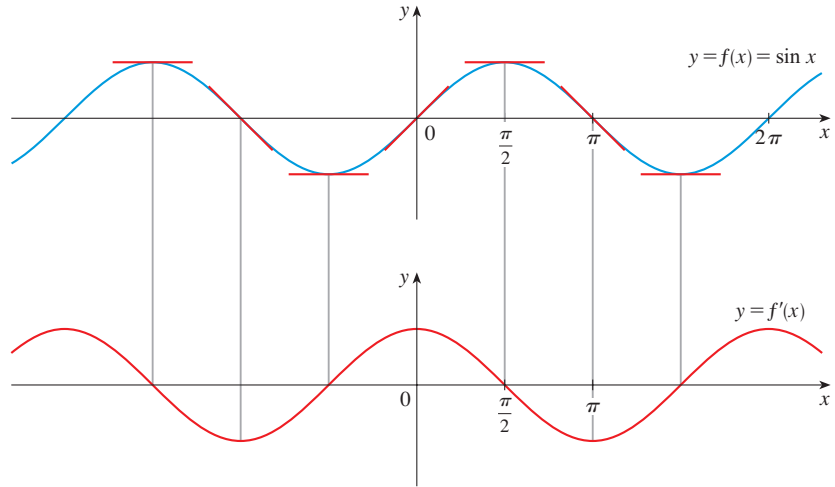


FIGURE 11

To prove that this is true we need to use two limits from Section 2.4 (see Equation 6 and Example 13 in that section):

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \quad \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$$

(7)

$$\frac{d}{dx}(\sin x) = \cos x$$

PROOF If $f(x) = \sin x$, then

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{\sin x \cos h - \sin x}{h} + \frac{\cos x \sin h}{h} \right] \\ &= \lim_{h \rightarrow 0} \left[\sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \left(\frac{\sin h}{h} \right) \right] \\ &= \lim_{h \rightarrow 0} \sin x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \lim_{h \rightarrow 0} \cos x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ &= (\sin x) \cdot 0 + (\cos x) \cdot 1 = \cos x \end{aligned}$$

We have used the addition formula for sine. See Appendix C.

Note that we regard x as a constant when computing a limit as $h \rightarrow 0$, so $\sin x$ and $\cos x$ are also constants.

Using the same methods as in the proof of Formula 7, one can prove (see Exercise 70) that

(8)

$$\frac{d}{dx}(\cos x) = -\sin x$$

EXAMPLE 12 | Differentiate $y = 3 \sin \theta + 4 \cos \theta$.**SOLUTION**

$$\frac{dy}{d\theta} = 3 \frac{d}{d\theta} (\sin \theta) + 4 \frac{d}{d\theta} (\cos \theta) = 3 \cos \theta - 4 \sin \theta$$

EXAMPLE 13 | Find the 27th derivative of $\cos x$.**SOLUTION** The first few derivatives of $f(x) = \cos x$ are as follows:

$$f'(x) = -\sin x$$

$$f''(x) = -\cos x$$

$$f'''(x) = \sin x$$

$$f^{(4)}(x) = \cos x$$

$$f^{(5)}(x) = -\sin x$$

Looking for a pattern, we see that the successive derivatives occur in a cycle of length 4 and, in particular, $f^{(n)}(x) = \cos x$ whenever n is a multiple of 4. Therefore

$$f^{(24)}(x) = \cos x$$

and, differentiating three more times, we have

$$f^{(27)}(x) = \sin x$$

EXERCISES 3.3

1. (a) How is the number e defined?
- (b) Use a calculator to estimate the values of the limits

$$\lim_{h \rightarrow 0} \frac{2.7^h - 1}{h} \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{2.8^h - 1}{h}$$

correct to two decimal places. What can you conclude about the value of e ?

2. (a) Sketch, by hand, the graph of the function $f(x) = e^x$, paying particular attention to how the graph crosses the y -axis. What fact allows you to do this?
- (b) What types of functions are $f(x) = e^x$ and $g(x) = x^e$? Compare the differentiation formulas for f and g .
- (c) Which of the two functions in part (b) grows more rapidly when x is large?

3–32 Differentiate the function.

3. $f(x) = 186.5$
5. $f(x) = 5x - 1$
7. $f(x) = x^3 - 4x + 6$
9. $f(x) = x - 3 \sin x$
11. $f(t) = \frac{1}{4}(t^4 + 8)$
4. $f(x) = \sqrt{30}$
6. $F(x) = \frac{3}{4}x^8$
8. $f(t) = \frac{1}{2}t^6 - 3t^4 + t$
10. $y = \sin t + \pi \cos t$
12. $h(x) = (x - 2)(2x + 3)$

$$13. A(s) = -\frac{12}{s^5}$$

$$15. g(t) = 2t^{-3/4}$$

$$17. y = 3e^x + \frac{4}{\sqrt[3]{x}}$$

$$19. F(x) = \left(\frac{1}{2}x\right)^5$$

$$21. y = \frac{x^2 + 4x + 3}{\sqrt{x}}$$

$$23. y = 4\pi^2$$

$$25. g(y) = \frac{A}{y^{10}} + B \cos y$$

$$27. f(x) = k(a - x)(b - x)$$

$$29. u = \sqrt[5]{t} + 4\sqrt{t^5}$$

$$31. G(y) = \frac{A}{y^{10}} + Be^y$$

$$14. B(y) = cy^{-6}$$

$$16. h(t) = \sqrt[4]{t} - 4e^t$$

$$18. y = \sqrt{x}(x - 1)$$

$$20. f(x) = \frac{x^2 - 3x + 1}{x^2}$$

$$22. g(u) = \sqrt{2}u + \sqrt{3}u$$

$$24. L(\theta) = \frac{\sin \theta}{2} + \frac{c}{\theta}$$

$$26. h(N) = rN \left(1 - \frac{N}{K}\right)$$

$$28. F(v) = ae^v + \frac{b}{v} + \frac{c}{v^2}$$

$$30. v = \left(\sqrt{x} + \frac{1}{\sqrt[3]{x}}\right)^2$$

$$32. y = e^{x+1} + 1$$

33–34 Find an equation of the tangent line to the curve at the given point.

33. $y = \sqrt[4]{x}$, $(1, 1)$

34. $y = x^4 + 2x^2 - x$, $(1, 2)$

35–38 Find equations of the tangent line and normal line to the curve at the given point.

35. $y = 6 \cos x$, $(\pi/3, 3)$ **36.** $y = x^2 - x^4$, $(1, 0)$

37. $y = x^4 + 2e^x$, $(0, 2)$ **38.** $y = (1 + 2x)^2$, $(1, 9)$

 **39–40** Find an equation of the tangent line to the curve at the given point. Illustrate by graphing the curve and the tangent line on the same screen.

39. $y = x + \sqrt{x}$, $(1, 2)$ **40.** $y = 3x^2 - x^3$, $(1, 2)$

 **41–42** Find $f'(x)$. Compare the graphs of f and f' and use them to explain why your answer is reasonable.

41. $f(x) = 3x^{15} - 5x^3 + 3$ **42.** $f(x) = x + \frac{1}{x}$

43–46 Find the first and second derivatives of the function.

43. $f(x) = x^4 - 3x^3 + 16x$ **44.** $G(r) = \sqrt{r} + \sqrt[3]{r}$

45. $g(t) = 2 \cos t - 3 \sin t$ **46.** $h(t) = \sqrt{t} + 5 \sin t$

47. Fish growth Biologists have proposed a cubic polynomial to model the length L of rock bass at age A :

$$L = 0.0155A^3 - 0.372A^2 + 3.95A + 1.21$$

where L is measured in inches and A in years. (See Exercise 1.2.27.) Calculate

$$\left. \frac{dL}{dA} \right|_{A=12}$$

and interpret your answer.

48. Rain forest biodiversity The number of tree species S in a given area A in the Pasoh Forest Reserve in Malaysia has been modeled by the power function

$$S(A) = 0.882A^{0.842}$$

where A is measured in square meters. Find $S'(100)$ and interpret your answer.

(Source: Adapted from K. Kochummen et al., “Floristic Composition of Pasoh Forest Reserve, a Lowland Rain Forest in Peninsular Malaysia,” *Journal of Tropical Forest Science* 3 (1991): 1–13.

49. Blood flow Refer to the law of laminar flow given in Example 9. Consider a blood vessel with radius 0.01 cm,

length 3 cm, pressure difference 3000 dynes/cm², and viscosity $\eta = 0.027$.

- Find the velocity of the blood along the center-line $r = 0$, at radius $r = 0.005$ cm, and at the wall $r = R = 0.01$ cm.
- Find the velocity gradient at $r = 0$, $r = 0.005$, and $r = 0.01$.
- Where is the velocity the greatest? Where is the velocity changing most?

50. Invasive species often display a wave of advance as they colonize new areas. Mathematical models based on random dispersal and reproduction have demonstrated that the speed with which such waves move is given by the expression $2\sqrt{Dr}$, where r is the reproductive rate of individuals and D is a parameter quantifying dispersal. Calculate the derivative of the wave speed with respect to the reproductive rate r and explain its meaning.

51. The position function of a particle is given by $s = t^3 - 4.5t^2 - 7t$, $t \geq 0$.

- Find the velocity and acceleration of the particle.
- When does the particle reach a velocity of 5 m/s?
- When is the acceleration 0? What is the significance of this value of t ?

52. If a ball is given a push so that it has an initial velocity of 5 m/s rolling down a certain inclined plane, then the distance it has rolled after t seconds is $s = 5t + 3t^2$.

- Find the velocity after 2 s.
- How long does it take for the velocity to reach 35 m/s?

53. (a) A company makes computer chips from square wafers of silicon. It wants to keep the side length of a wafer very close to 15 mm and it wants to know how the area $A(x)$ of a wafer changes when the side length x changes. Find $A'(15)$ and explain its meaning in this situation.

- Show that the rate of change of the area of a square with respect to its side length is half its perimeter. Try to explain geometrically why this is true by drawing a square whose side length x is increased by an amount Δx . How can you approximate the resulting change in area ΔA if Δx is small?

54. (a) Sodium chlorate crystals are easy to grow in the shape of cubes by allowing a solution of water and sodium chlorate to evaporate slowly. If V is the volume of such a cube with side length x , calculate dV/dx when $x = 3$ mm and explain its meaning.

- Show that the rate of change of the volume of a cube with respect to its edge length is equal to half the surface area of the cube. Explain geometrically why this result is true by arguing by analogy with Exercise 53(b).

55. (a) Find the average rate of change of the area of a circle with respect to its radius r as r changes from

- 2 to 3 (ii) 2 to 2.5 (iii) 2 to 2.1
- Find the instantaneous rate of change when $r = 2$.

- (c) Show that the rate of change of the area of a circle with respect to its radius (at any r) is equal to the circumference of the circle. Try to explain geometrically why this is true by drawing a circle whose radius is increased by an amount Δr . How can you approximate the resulting change in area ΔA if Δr is small?
56. (a) **Cell growth** The volume of a growing spherical cell is $V = \frac{4}{3}\pi r^3$, where the radius r is measured in micrometers ($1\ \mu\text{m} = 10^{-6}\ \text{m}$). Find the average rate of change of V with respect to r when r changes from
(i) 5 to $8\ \mu\text{m}$ (ii) 5 to $6\ \mu\text{m}$ (iii) 5 to $5.1\ \mu\text{m}$
(b) Find the instantaneous rate of change of V with respect to r when $r = 5\ \mu\text{m}$.
(c) Show that the rate of change of the volume of a cell with respect to its radius is equal to its surface area ($S = 4\pi r^2$). Explain geometrically why this result is true. Argue by analogy with Exercise 55(c).
57. Find $\frac{d^{99}}{dx^{99}}(\sin x)$.
58. Find the n th derivative of each function by calculating the first few derivatives and observing the pattern that occurs.
(a) $f(x) = x^n$ (b) $f(x) = 1/x$
59. For what values of x does the graph of $f(x) = x + 2\sin x$ have a horizontal tangent?
60. For what values of x does the graph of $f(x) = x^3 + 3x^2 + x + 3$ have a horizontal tangent?
61. Show that the curve $y = 6x^3 + 5x - 3$ has no tangent line with slope 4.
62. Find an equation of the tangent line to the curve $y = x\sqrt{x}$ that is parallel to the line $y = 1 + 3x$.
63. Find equations of both lines that are tangent to the curve $y = 1 + x^3$ and parallel to the line $12x - y = 1$.
-  64. At what point on the curve $y = 1 + 2e^x - 3x$ is the tangent line parallel to the line $3x - y = 5$? Illustrate by graphing the curve and both lines.
65. Find an equation of the normal line to the parabola $y = x^2 - 5x + 4$ that is parallel to the line $x - 3y = 5$.
66. Where does the normal line to the parabola $y = x - x^2$ at the point $(1, 0)$ intersect the parabola a second time? Illustrate with a sketch.
67. Draw a diagram to show that there are two tangent lines to the parabola $y = x^2$ that pass through the point $(0, -4)$. Find the coordinates of the points where these tangent lines intersect the parabola.
68. (a) Find equations of both lines through the point $(2, -3)$ that are tangent to the parabola $y = x^2 + x$.
(b) Show that there is no line through the point $(2, 7)$ that is tangent to the parabola. Then draw a diagram to see why.
69. Use the definition of a derivative to show that if $f(x) = 1/x$, then $f'(x) = -1/x^2$. (This proves the Power Rule for the case $n = -1$.)
70. Prove, using the definition of derivative, that if $f(x) = \cos x$, then $f'(x) = -\sin x$.
71. Find the parabola with equation $y = ax^2 + bx$ whose tangent line at $(1, 1)$ has equation $y = 3x - 2$.
72. Find the value of c such that the line $y = \frac{3}{2}x + 6$ is tangent to the curve $y = c\sqrt{x}$.
73. For what values of a and b is the line $2x + y = b$ tangent to the parabola $y = ax^2$ when $x = 2$?
74. A tangent line is drawn to the hyperbola $xy = c$ at a point P .
(a) Show that the midpoint of the line segment cut from this tangent line by the coordinate axes is P .
(b) Show that the triangle formed by the tangent line and the coordinate axes always has the same area, no matter where P is located on the hyperbola.
75. Evaluate $\lim_{x \rightarrow 1} \frac{x^{1000} - 1}{x - 1}$.
76. Draw a diagram showing two perpendicular lines that intersect on the y -axis and are both tangent to the parabola $y = x^2$. Where do these lines intersect?

3.4 The Product and Quotient Rules

The formulas of this section enable us to differentiate new functions formed from old functions by multiplication or division.

■ The Product Rule



By analogy with the Sum and Difference Rules, one might be tempted to guess, as Leibniz did three centuries ago, that the derivative of a product is the product of the derivatives. We can see, however, that this guess is wrong by looking at a particular example. Let $f(x) = x$ and $g(x) = x^2$. Then the Power Rule gives $f'(x) = 1$ and $g'(x) = 2x$. But

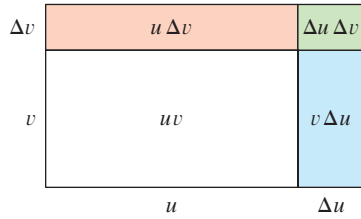


FIGURE 1
The geometry of the Product Rule

$(fg)(x) = x^3$, so $(fg)'(x) = 3x^2$. Thus $(fg)' \neq f'g'$. The correct formula was discovered by Leibniz (soon after his false start) and is called the Product Rule.

Before stating the Product Rule, let's see how we might discover it. We start by assuming that $u = f(x)$ and $v = g(x)$ are both positive differentiable functions. Then we can interpret the product uv as an area of a rectangle (see Figure 1). If x changes by an amount Δx , then the corresponding changes in u and v are

$$\Delta u = f(x + \Delta x) - f(x) \quad \Delta v = g(x + \Delta x) - g(x)$$

and the new value of the product, $(u + \Delta u)(v + \Delta v)$, can be interpreted as the area of the large rectangle in Figure 1 (provided that Δu and Δv happen to be positive).

The change in the area of the rectangle is

$$\begin{aligned} (1) \quad \Delta(uv) &= (u + \Delta u)(v + \Delta v) - uv = u \Delta v + v \Delta u + \Delta u \Delta v \\ &= \text{the sum of the three shaded areas} \end{aligned}$$

If we divide by Δx , we get

$$\frac{\Delta(uv)}{\Delta x} = u \frac{\Delta v}{\Delta x} + v \frac{\Delta u}{\Delta x} + \Delta u \frac{\Delta v}{\Delta x}$$

Recall that in Leibniz notation the definition of a derivative can be written as

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

If we now let $\Delta x \rightarrow 0$, we get the derivative of uv :

$$\begin{aligned} \frac{d}{dx}(uv) &= \lim_{\Delta x \rightarrow 0} \frac{\Delta(uv)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left(u \frac{\Delta v}{\Delta x} + v \frac{\Delta u}{\Delta x} + \Delta u \frac{\Delta v}{\Delta x} \right) \\ &= u \lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta x} + v \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} + \left(\lim_{\Delta x \rightarrow 0} \Delta u \right) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta x} \right) \\ &= u \frac{dv}{dx} + v \frac{du}{dx} + 0 \cdot \frac{dv}{dx} \end{aligned}$$

$$(2) \quad \frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

(Notice that $\Delta u \rightarrow 0$ as $\Delta x \rightarrow 0$ since f is differentiable and therefore continuous.)

Although we started by assuming (for the geometric interpretation) that all the quantities are positive, we notice that Equation 1 is always true. (The algebra is valid whether u , v , Δu , and Δv are positive or negative.) So we have proved Equation 2, known as the Product Rule, for all differentiable functions u and v .

In prime notation:

$$(fg)' = fg' + gf'$$

The Product Rule If f and g are both differentiable, then

$$\frac{d}{dx}[f(x)g(x)] = f(x) \frac{d}{dx}[g(x)] + g(x) \frac{d}{dx}[f(x)]$$

In words, the Product Rule says that *the derivative of a product of two functions is the first function times the derivative of the second function plus the second function times the derivative of the first function.*

Figure 2 shows the graphs of the function of Example 1 and its derivative. Notice that $y' = 0$ whenever y has a horizontal tangent.

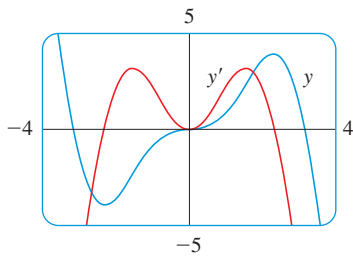


FIGURE 2

In Example 2, a and b are constants. It is customary in mathematics to use letters near the beginning of the alphabet to represent constants and letters near the end of the alphabet to represent variables.

EXAMPLE 1 | Differentiate $y = x^2 \sin x$.

SOLUTION Using the Product Rule, we have

$$\begin{aligned}\frac{dy}{dx} &= x^2 \frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(x^2) \\ &= x^2 \cos x + 2x \sin x\end{aligned}$$

EXAMPLE 2 | Differentiate the function $f(t) = \sqrt{t}(a + bt)$.

SOLUTION 1 Using the Product Rule, we have

$$\begin{aligned}f'(t) &= \sqrt{t} \frac{d}{dt}(a + bt) + (a + bt) \frac{d}{dt}(\sqrt{t}) \\ &= \sqrt{t} \cdot b + (a + bt) \cdot \frac{1}{2}t^{-1/2} \\ &= b\sqrt{t} + \frac{a + bt}{2\sqrt{t}} = \frac{a + 3bt}{2\sqrt{t}}\end{aligned}$$

SOLUTION 2 If we first use the laws of exponents to rewrite $f(t)$, then we can proceed directly without using the Product Rule.

$$\begin{aligned}f(t) &= a\sqrt{t} + bt\sqrt{t} = at^{1/2} + bt^{3/2} \\ f'(t) &= \frac{1}{2}at^{-1/2} + \frac{3}{2}bt^{1/2}\end{aligned}$$

which is equivalent to the answer given in Solution 1.

Example 2 shows that it is sometimes easier to simplify a product of functions before differentiating than to use the Product Rule. In Example 1, however, the Product Rule is the only possible method.

EXAMPLE 3 | If $h(x) = xg(x)$ and it is known that $g(3) = 5$ and $g'(3) = 2$, find $h'(3)$.

SOLUTION Applying the Product Rule, we get

$$\begin{aligned}h'(x) &= \frac{d}{dx}[xg(x)] \\ &= x \frac{d}{dx}[g(x)] + g(x) \frac{d}{dx}[x] \\ &= xg'(x) + g(x)\end{aligned}$$

Therefore

$$h'(3) = 3g'(3) + g(3) = 3 \cdot 2 + 5 = 11$$

■ The Quotient Rule

We find a rule for differentiating the quotient of two differentiable functions $u = f(x)$ and $v = g(x)$ in much the same way that we found the Product Rule. If x , u , and v change

by amounts Δx , Δu , and Δv , then the corresponding change in the quotient u/v is

$$\begin{aligned}\Delta\left(\frac{u}{v}\right) &= \frac{u + \Delta u}{v + \Delta v} - \frac{u}{v} = \frac{(u + \Delta u)v - u(v + \Delta v)}{v(v + \Delta v)} \\ &= \frac{v\Delta u - u\Delta v}{v(v + \Delta v)}\end{aligned}$$

so

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \lim_{\Delta x \rightarrow 0} \frac{\Delta(u/v)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{v \frac{\Delta u}{\Delta x} - u \frac{\Delta v}{\Delta x}}{v(v + \Delta v)}$$

As $\Delta x \rightarrow 0$, $\Delta v \rightarrow 0$ also, because $v = g(x)$ is differentiable and therefore continuous. Thus, using the Limit Laws, we get

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} - u \lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta x}}{v \lim_{\Delta x \rightarrow 0} (v + \Delta v)} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

In prime notation:

$$\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$$

The Quotient Rule If f and g are differentiable, then

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} [f(x)] - f(x) \frac{d}{dx} [g(x)]}{[g(x)]^2}$$

In words, the Quotient Rule says that the *derivative of a quotient is the denominator times the derivative of the numerator minus the numerator times the derivative of the denominator, all divided by the square of the denominator.*

The Quotient Rule and the other differentiation formulas enable us to compute the derivative of any rational function, as the next example illustrates.

We can use a graphing device to check that the answer to Example 4 is plausible. Figure 3 shows the graphs of the function of Example 4 and its derivative. Notice that when y grows rapidly (near -2), y' is large. And when y grows slowly, y' is near 0.

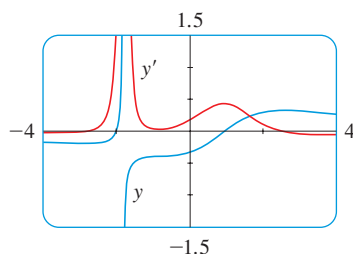


FIGURE 3

EXAMPLE 4 | Let $y = \frac{x^2 + x - 2}{x^3 + 6}$. Then

$$\begin{aligned}y' &= \frac{(x^3 + 6) \frac{d}{dx} (x^2 + x - 2) - (x^2 + x - 2) \frac{d}{dx} (x^3 + 6)}{(x^3 + 6)^2} \\ &= \frac{(x^3 + 6)(2x + 1) - (x^2 + x - 2)(3x^2)}{(x^3 + 6)^2} \\ &= \frac{(2x^4 + x^3 + 12x + 6) - (3x^4 + 3x^3 - 6x^2)}{(x^3 + 6)^2} \\ &= \frac{-x^4 - 2x^3 + 6x^2 + 12x + 6}{(x^3 + 6)^2}\end{aligned}$$

EXAMPLE 5 | **BB** The Monod growth function Monod modeled the per capita growth rate R of *Escherichia coli* bacteria by the function

$$R(N) = \frac{SN}{c + N}$$

where N is the concentration of the nutrient, S is its saturation level, and c is a positive constant. Calculate dR/dN and interpret it.

SOLUTION Using the Quotient Rule, we have

$$\begin{aligned} \frac{dR}{dN} &= \frac{(c + N) \frac{d}{dN}(SN) - SN \frac{d}{dN}(c + N)}{(c + N)^2} \\ &= \frac{(c + N)S - SN \cdot 1}{(c + N)^2} = \frac{cS}{(c + N)^2} \end{aligned}$$

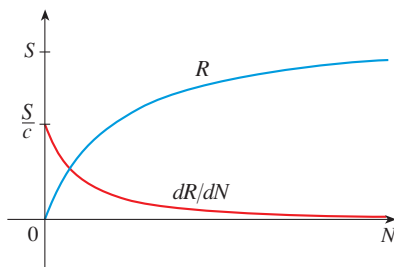


FIGURE 4

The derivative dR/dN is the rate of change of the growth rate R with respect to the concentration of the nutrient. From the expression for dR/dN we see that it is always positive, which means that R is an increasing function of N . But as the concentration becomes larger, dR/dN becomes smaller because of the denominator $(c + N)^2$. So

$$\frac{dR}{dN} \rightarrow 0 \quad \text{as} \quad N \rightarrow \infty$$

and this means that the graph of R becomes flatter as N gets larger. (See Figure 4.)

EXAMPLE 6 | Find an equation of the tangent line to the curve $y = e^x/(1 + x^2)$ at the point $(1, \frac{1}{2}e)$.

SOLUTION According to the Quotient Rule, we have

$$\begin{aligned} \frac{dy}{dx} &= \frac{(1 + x^2) \frac{d}{dx}(e^x) - e^x \frac{d}{dx}(1 + x^2)}{(1 + x^2)^2} \\ &= \frac{(1 + x^2)e^x - e^x(2x)}{(1 + x^2)^2} = \frac{e^x(1 - 2x + x^2)}{(1 + x^2)^2} \\ &= \frac{e^x(1 - x)^2}{(1 + x^2)^2} \end{aligned}$$

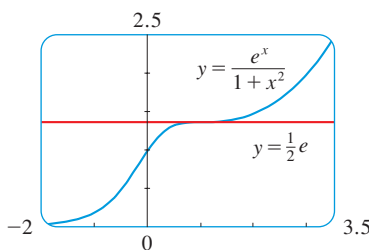


FIGURE 5

So the slope of the tangent line at $(1, \frac{1}{2}e)$ is

$$\left. \frac{dy}{dx} \right|_{x=1} = 0$$

This means that the tangent line at $(1, \frac{1}{2}e)$ is horizontal and its equation is $y = \frac{1}{2}e$. [See Figure 5. Notice that the function is increasing and crosses its tangent line at $(1, \frac{1}{2}e)$.]

NOTE Don't use the Quotient Rule *every* time you see a quotient. Sometimes it's easier to rewrite a quotient first to put it in a form that is simpler for the purpose of differentiation. For instance, although it is possible to differentiate the function

$$F(x) = \frac{3x^2 + 2\sqrt{x}}{x}$$

using the Quotient Rule, it is much easier to perform the division first and write the function as

$$F(x) = 3x + 2x^{-1/2}$$

before differentiating.

We summarize the differentiation formulas we have learned so far as follows.

Table of Differentiation Formulas

$\frac{d}{dx}(c) = 0$	$\frac{d}{dx}(x^n) = nx^{n-1}$	$\frac{d}{dx}(e^x) = e^x$
$(cf)' = cf'$	$(f + g)' = f' + g'$	$(f - g)' = f' - g'$
$(fg)' = fg' + gf'$	$\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$	

■ Trigonometric Functions

Knowing the derivatives of the sine and cosine functions, we can use the Quotient Rule to find the derivative of the tangent function:

$$\begin{aligned}
 \frac{d}{dx}(\tan x) &= \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) \\
 &= \frac{\cos x \frac{d}{dx}(\sin x) - \sin x \frac{d}{dx}(\cos x)}{\cos^2 x} \\
 &= \frac{\cos x \cdot \cos x - \sin x(-\sin x)}{\cos^2 x} \\
 &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\
 &= \frac{1}{\cos^2 x} = \sec^2 x
 \end{aligned}$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

The derivatives of the remaining trigonometric functions, csc, sec, and cot, can also be found easily using the Quotient Rule (see Exercises 45–47). We collect all the differentiation formulas for trigonometric functions in the following table. Remember that they are valid only when x is measured in radians.